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UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 21

HYPOTHESIS TESTING

SUGGESTED TIME: 4 PERIODS

IN THIS CHAPTER

This chapter introduces significance (hypothesis) testing: how to set up null and alternative hypotheses, choose significance levels, compute test statistics and p-values, and draw conclusions. Tests are developed for a probability (proportion), the mean of a Poisson distribution, a population mean, and the difference between two means using the Central Limit Theorem.

- By the end of this chapter I can state a null hypothesis H0 and an alternative hypothesis H1 appropriate to a given situation.
- By the end of this chapter I can distinguish one-tailed from two-tailed tests and choose the correct form of H1.
- By the end of this chapter I can define the significance level, the critical region, Type I and Type II errors.
- By the end of this chapter I can test a hypothetical value of a probability p using the binomial distribution.
- By the end of this chapter I can test a hypothetical value of the mean of a Poisson distribution.
- By the end of this chapter I can test a population mean using a normal or z-test (known variance or large sample).

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METHODS
and worked examples

EXERCISES
exam style + solutions

SUMMARY SHEET
the essentials to remember

CONTENTS

1. Foundations of Significance Testing: Hypotheses, Errors and Critical Regions
2. Testing a Probability and the Mean of a Poisson Distribution
3. Testing a Population Mean
4. Testing the Difference Between Two Means (Large Samples)
5. Integration activity
6. Methods and worked examples
7. Exercises
8. Solutions to the exercises
9. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can state a null hypothesis H0 and an alternative hypothesis H1 appropriate to a given situation.
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- By the end of this chapter I can define the significance level, the critical region, Type I and Type II errors.
- By the end of this chapter I can test a hypothetical value of a probability p using the binomial distribution.
- By the end of this chapter I can test a hypothetical value of the mean of a Poisson distribution.
- By the end of this chapter I can test a population mean using a normal or z-test (known variance or large sample).
- By the end of this chapter I can compute and interpret a p-value and compare it with the significance level.
- By the end of this chapter I can test the difference between two population means for large samples using the Central Limit Theorem.
- By the end of this chapter I can write a clear conclusion in context, linking the statistical decision to the original claim.

PREREQUISITES

- Binomial, Poisson and Normal distributions and their means and variances (Lower Sixth and Upper Sixth probability chapters).
- The Central Limit Theorem and the sampling distribution of the sample mean.
- Standardisation of a normal variable and use of standard normal tables.
- Confidence intervals for a mean (useful link with two-tailed tests).
- Basic probability computations: cumulative probabilities $P(X \leq k)$ and $P(X \geq k)$.

1 FOUNDATIONS OF SIGNIFICANCE TESTING: HYPOTHESES, ERRORS AND CRITICAL REGIONS

1.1 The Logic of Significance Testing

A Yaoundé bottling company claims its sachets of "pure water" contain 500 ml on average, but a consumer association suspects under-filling. A mobile money agent in Douala claims that only 2 % of transactions fail, yet customers complain of frequent failures. A health officer in Bamenda wants to know whether the mean number of malaria cases per week at a clinic has really increased after a flood. In each case, a **claim** is made about a **population**, a **sample** is taken, and a **decision** must be reached under **uncertainty**. **Hypothesis testing** gives us the rigorous statistical machinery to decide whether the sample evidence is strong enough to reject the claim.

The core logic is the same as a courtroom trial. The accused is presumed innocent; the prosecution must produce evidence so strong that "innocence" becomes untenable. In statistics, the claim (the status quo) plays the role of "innocent":

1. **Assume the claim is true.** We take the claim about the population as a working assumption.
2. **Collect sample evidence.** We observe data from a random sample.
3. **Ask: how likely is this evidence if the claim were true?** We compute the probability of obtaining a result at least as extreme as the one observed, under the assumption that the claim holds.
4. **Decide.** If that probability is very small, the claim and the data do not sit well together, so we reject the claim. Otherwise we say the evidence is insufficient to reject it.

DID YOU KNOW? **Why "too unlikely" matters**

Suppose a coin is claimed to be fair. You toss it 20 times and get 19 heads. A fair coin could do this — the probability is about 0.00002 — but it is so unlikely that the reasonable conclusion is that the coin is not fair. Significance testing formalises exactly this reasoning, replacing "so unlikely" by a precise threshold.

1.2 The Null and Alternative Hypotheses

DEFINITION **Null and alternative hypotheses**

The **null hypothesis**, written H_0, is the claim or status quo about a population parameter. It is the statement we assume true while carrying out the test. The **alternative hypothesis**, written H_1, is what we suspect is true instead; it is accepted only if the sample evidence against H_0 is strong enough.

WARNING **H_0 always contains equality**

The null hypothesis always specifies a single value with an equality sign:

$$H_0 : p = p_0, \qquad H_0 : \mu = \mu_0, \qquad H_0 : \lambda = \lambda_0.$$

Never write $H_0 : p > p_0$ or $H_0 : \mu \neq \mu_0$. The equality is essential because the whole calculation of probabilities is performed assuming the parameter equals the claimed value.

One-tailed versus two-tailed tests.

The form of H_1 determines the tail of the test:

Type of test	Form of H_1	Typical wording
Lower one-tailed	$H_1 : p < p_0$	"decreased", "less than", "under-filled"
Upper one-tailed	$H_1 : p > p_0$	"increased", "more than", "improved"
Two-tailed	$H_1 : p \neq p_0$	"changed", "different from", "not equal to"

WARNING **The tail is fixed by H_1, not by the data**

A frequent and serious error is to look at the sample first and then choose the tail that makes the result significant. This is dishonest statistics: the hypotheses must be stated before examining the data, and the tail is dictated by the wording of the suspicion in the problem. If the consumer association suspects under-filling, the test is lower-tailed even if the sample mean happens to exceed 500 ml.

EXEMPLE **Stating hypotheses in context**

For the three opening situations:

- Bottling company: $H_0 : \mu = 500$ against $H_1 : \mu < 500$ (lower one-tailed, since under-filling is suspected).
- Mobile money agent: $H_0 : p = 0.02$ against $H_1 : p > 0.02$ (upper one-tailed, since customers suspect more failures).
- Health officer: $H_0 : \lambda = \lambda_0$ against $H_1 : \lambda > \lambda_0$ (upper one-tailed, since an increase is suspected).

1.3 The Test Statistic and its Sampling Distribution

DEFINITION **Test statistic**

A **test statistic** is a quantity computed from the sample whose value measures how compatible the data are with H_0. Its **sampling distribution under H_0** is the distribution it would have if H_0 were true; this distribution is what allows us to compute probabilities.

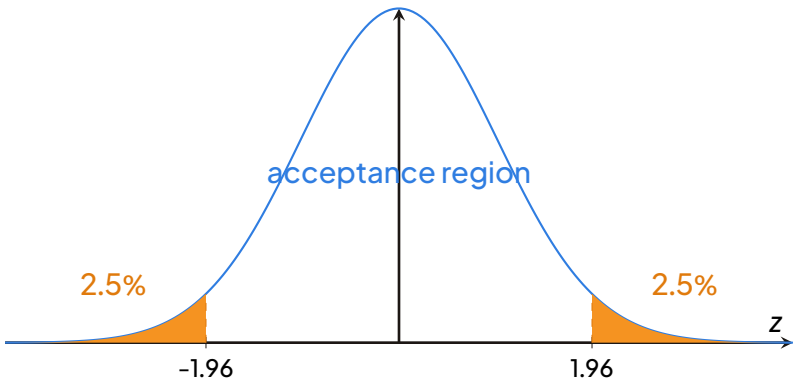
For example, if a coin is claimed fair ($H_0 : p = 0.5$) and we toss it $n = 20$ times, the natural test statistic is X, the number of heads, and under H_0 we know exactly that $X \sim \mathrm{B}(20, 0.5)$. If a population mean is claimed to be μ_0 and the population variance σ^2 is known, then under H_0 the sample mean satisfies $\bar{X} \sim \mathrm{N}\left(\mu_0, \frac{\sigma^2}{n}\right)$. Choosing the right test statistic and its distribution under H_0 is the technical heart of every test in this chapter.

1.4 Significance Level, Critical Region and Critical Values

DEFINITION **Significance level and critical region**

The **significance level** α of a test is the probability, fixed in advance, of rejecting H_0 when H_0 is actually true. Common choices are $\alpha = 5\%$ and $\alpha = 1\%$. The **critical region** (or rejection region) is the set of values of the test statistic that lead to rejection of H_0; its total probability under H_0 is (at most) α. The complementary set is the **acceptance region**. The boundary values are the **critical values**.

For a two-tailed test on a standard normal statistic at the 5% level, the critical values are ±1.96: we reject H_0 when $|z| > 1.96$. For one-tailed tests at 5%, the single critical value is −1.645 (lower tail) or +1.645 (upper tail).



Two-tailed critical region for a standard normal test statistic at $\alpha = 5\%$: reject H_0 when $|z| > 1.96$.

1.5 The p-value Approach

DEFINITION **p-value**

The **p-value** is the probability, computed under H_0, of obtaining a result at least as extreme (in the direction of H_1) as the one actually observed. The decision rule is:

$$\text{reject } H_0 \iff p\text{-value} < \alpha.$$

"At least as extreme" means: for an upper-tailed test, $P(X \geq x_{\text{obs}})$; for a lower-tailed test, $P(X \leq x_{\text{obs}})$; for a two-tailed test, the probability in both tails at least as far from the centre. The p-value quantifies the strength of the evidence: the smaller it is, the stronger the case against H_0.

WORKED EXAMPLE 1 **A first complete test: the mobile money agent**

A mobile money agent in Douala claims that the proportion of failed transactions is 2 %. A customer watchdog records $n = 100$ transactions and observes 6 failures. Test at the 5% level whether the failure rate exceeds 2 %.

SOLUTION

Solution. Let p be the true proportion of failed transactions.
Step 1 — Hypotheses.

$$H_0 : p = 0.02 \qquad H_1 : p > 0.02 \quad \text{(upper one-tailed)}.$$

Step 2 — Level. $\alpha = 5\%$.
Step 3 — Test statistic and distribution under H_0. Let X be the number of failures in 100 transactions. Under H_0, $X \sim \text{B}(100, 0.02)$. Since $np = 2$ is small, we use the Poisson approximation $X \approx \text{Po}(2)$.
Step 4 — p-value. The observed value is $x = 6$:

$$p\text{-value} = P(X \geq 6) = 1 - P(X \leq 5) = 1 - e^{-2}\sum_{k=0}^{5}\frac{2^k}{k!}.$$

Computing the sum: $e^{-2}\left(1 + 2 + 2 + \frac{4}{3} + \frac{2}{3} + \frac{4}{15}\right) = e^{-2}(7.2667) \approx 0.9834$.

$$p\text{-value} \approx 1 - 0.9834 = 0.0166.$$

Step 5 — Conclusion. Since $0.0166 < 0.05$, we reject H_0 at the 5% level. There is sufficient evidence that the proportion of failed transactions exceeds 2 %.

1.6 Type I and Type II Errors

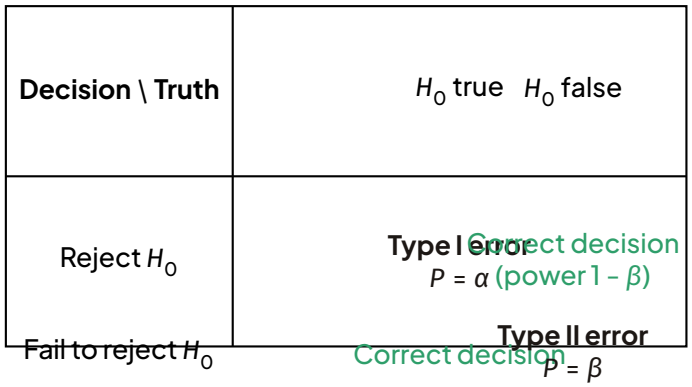
Because our decision is based on a sample, it can be wrong in two distinct ways.

DEFINITION **Type I and Type II errors**

A **Type I error** occurs when we reject H_0 although H_0 is true. By construction,

$$P(\text{Type I error}) = \alpha.$$

A **Type II error** occurs when we fail to reject H_0 although H_0 is false. Its probability is usually denoted β and depends on the true value of the parameter.



The two possible errors: rejecting a true H_0 (Type I, probability α) or failing to reject a false H_0 (Type II, probability β).

Consequences in context.

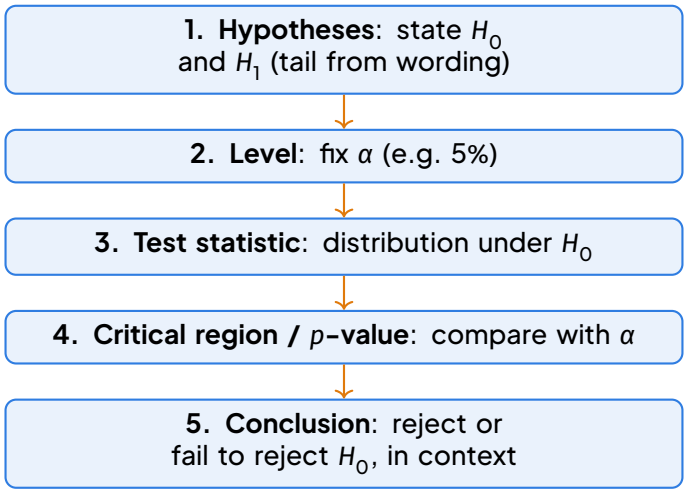
For the bottling company, a Type I error means wrongly accusing an honest company of under-filling (damaging its reputation); a Type II error means letting a dishonest company continue cheating consumers. Decreasing α (say from 5% to 1%) makes Type I errors rarer but, for a fixed sample size, makes Type II errors more likely — there is a trade-off, which is why α must be chosen before the test.

1.7 The Five-Step Procedure and the Conclusion Template

METHOD **The five steps of a significance test**

1. State the hypotheses H_0 and H_1, defining the parameter in context and choosing the tail from the wording.
2. State the significance level α (e.g. 5%).
3. Identify the test statistic and its sampling distribution under H_0.
4. Compute the critical region or the p-value and compare with α (or with the observed value).

5. Conclude in context: reject or fail to reject H_0, and interpret the decision in the language of the problem.



The five-step significance testing procedure used throughout this chapter.

KEY POINTS **Standard conclusion template**

Every conclusion must contain three ingredients: the comparison, the decision, and the context.

> "Since p-value = ... < α = ..., we reject H_0 at the ... % level. There is sufficient evidence that [claim of H_1 in context]."

or

> "Since p-value = ... > α = ..., we do not reject H_0 at the ... % level. There is insufficient evidence that [claim of H_1 in context]."

WARNING **Never "prove" H_0**

A significance test can never prove that H_0 is true. When the evidence is weak, the correct statement is "we fail to reject H_0" or "there is insufficient evidence against H_0" — never "we accept that H_0 is true". Absence of evidence is not evidence of absence: a small sample may simply lack the power to detect a real difference. In the courtroom analogy, "not guilty" does not mean "proven innocent".

EXERCISE 1 **Stating hypotheses and errors**

A clinic in Bamenda records a long-run mean of 12 malaria cases per week. After a flood, the health officer suspects the mean has increased. In a sample of weeks she will test the claim.

1. State H_0 and H_1, justifying the tail.
2. Describe, in context, what a Type I error and a Type II error would mean.
3. The officer computes a p-value of 0.032. State her conclusion at the 5% level, using the standard template.

SOLUTION **Exercise 1**

1. Let λ be the true mean number of cases per week. Since an increase is suspected, the test is upper one-tailed: $H_0 : \lambda = 12$ against $H_1 : \lambda > 12$.
2. Type I error: concluding that malaria cases have increased when in fact the mean is still 12 — this might trigger unnecessary emergency spending. Type II error: failing to detect a genuine increase — the clinic remains under-prepared while cases rise.
3. Since p-value $= 0.032 < 0.05$, we reject H_0 at the 5% level. There is sufficient evidence that the mean number of malaria cases per week has increased above 12.

2 TESTING A PROBABILITY AND THE MEAN OF A POISSON DISTRIBUTION

2.1 Testing a Binomial Proportion p

When the claim concerns a population proportion p, the natural test statistic is the number of successes X in a random sample of size n. Under $H_0 : p = p_0$, we have exactly $X \sim \mathrm{B}(n, p_0)$.

PROPERTY **Exact binomial test**

For $H_0 : p = p_0$ against $H_1 : p < p_0$, $p > p_0$ or $p \neq p_0$:

- Test statistic: $X \sim \mathrm{B}(n, p_0)$ under H_0.
- p-value: computed directly from the binomial distribution.
- Critical region: values of X with cumulative probability $\leq \alpha$ in the relevant tail(s).

WARNING **Discreteness of the binomial**

Because the binomial distribution is discrete, the actual significance level (the probability of the critical region) is usually less than the nominal α. We always choose the largest critical region whose probability does not exceed α.

EXEMPLE **Exact binomial test: lower tail**

A seed merchant claims that 80% of his maize seeds germinate. A farmer tests $n = 20$ seeds and only 12 germinate. Test at the 5% level whether the germination rate is less than 80%.

- $H_0 : p = 0.8$, $H_1 : p < 0.8$ (lower one-tailed), $\alpha = 0.05$.
- Under H_0, $X \sim \mathrm{B}(20, 0.8)$. Observed $x = 12$.
- p-value $= P(X \le 12) = \sum_{k=0}^{12} \binom{20}{k} 0.8^k 0.2^{20-k}$.
- Using tables or calculator: $P(X \le 12) \approx 0.0321$.
- Since $0.0321 < 0.05$, reject H_0. There is sufficient evidence that the germination rate is below 80%.

PROPERTY **Normal approximation to the binomial**

If n is large enough that $np_0 > 5$ and $n(1-p_0) > 5$, we may approximate $X \sim \mathrm{B}(n, p_0)$ by $X \approx \mathrm{N}(np_0, np_0(1-p_0))$. The test statistic becomes

$$Z = \frac{X - np_0}{\sqrt{np_0(1-p_0)}} \sim \mathrm{N}(0,1) \quad \text{under } H_0.$$

A continuity correction is recommended: for $P(X \ge x)$ use $P(X > x - 0.5)$; for $P(X \le x)$ use $P(X < x + 0.5)$.

WORKED EXAMPLE 2 **Normal approximation: two-tailed test**

A political party claims it has the support of 50% of voters. In a random sample of 400 voters, 220 support the party. Test at the 5% level whether the support differs from 50%.

SOLUTION

Solution.

- $H_0 : p = 0.5, H_1 : p \neq 0.5$ (two-tailed), $\alpha = 0.05$.
- $n = 400, np_0 = 200 > 5, n(1 - p_0) = 200 > 5 \Rightarrow$ normal approximation valid.
- Under H_0: $X \approx N(200, 100)$. Observed $x = 220$.
- With continuity correction: $P(X \geq 220) \approx P\left(Z > \frac{219.5-200}{10}\right) = P(Z > 1.95)$.
- Two-tailed p-value $= 2 \times P(Z > 1.95) = 2 \times 0.0256 = 0.0512$.
- Since $0.0512 > 0.05$, do not reject H_0. Insufficient evidence that support differs from 50%.

2.2 Testing the Mean λ of a Poisson Distribution

When the claim concerns the mean λ of a Poisson distribution, the test statistic is the total number of events X observed over a given interval (time, area, volume). Under $H_0 : \lambda = \lambda_0$, if we observe over t units, then $X \sim \text{Po}(t\lambda_0)$.

PROPERTY **Exact Poisson test**

For $H_0 : \lambda = \lambda_0$ against $H_1 : \lambda < \lambda_0, \lambda > \lambda_0$ or $\lambda \neq \lambda_0$:

- Test statistic: $X \sim \text{Po}(t\lambda_0)$ under H_0, where t is the number of units observed.
- p-value: computed directly from the Poisson distribution.
- Critical region: values of X with cumulative probability $\leq \alpha$ in the relevant tail(s).

EXEMPLE **Exact Poisson test: upper tail**

A call centre receives a mean of 8 calls per hour. After an advertising campaign, the manager suspects the rate has increased. In a 2-hour period, 22 calls are received. Test at the 5% level.

- $H_0 : \lambda = 8, H_1 : \lambda > 8$ (upper one-tailed), $\alpha = 0.05$.
- Over $t = 2$ hours, under H_0: $X \sim \text{Po}(16)$. Observed $x = 22$.
- p-value $= P(X \geq 22) = 1 - P(X \leq 21)$.
- From Poisson tables or calculator: $P(X \leq 21) \approx 0.9170$, so p-value ≈ 0.0830.
- Since $0.0830 > 0.05$, do not reject H_0. Insufficient evidence that the call rate has increased.

PROPERTY **Normal approximation to the Poisson**

If $\lambda_0 t > 15$ (some texts use > 10), we may approximate $X \sim \text{Po}(\lambda_0 t)$ by $X \approx \text{N}(\lambda_0 t, \lambda_0 t)$. The test statistic becomes

$$Z = \frac{X - \lambda_0 t}{\sqrt{\lambda_0 t}} \sim \text{N}(0,1) \quad \text{under } H_0.$$

A continuity correction is recommended.

WORKED EXAMPLE 3 **Normal approximation to Poisson: two-tailed**

A factory produces defects at a mean rate of 25 per day. After a new machine is installed, a sample of 10 days yields a total of 280 defects. Test at the 1% level whether the defect rate has changed.

SOLUTION

Solution.

- $H_0 : \lambda = 25$, $H_1 : \lambda \neq 25$ (two-tailed), $\alpha = 0.01$.
- Over $t = 10$ days, under H_0: $X \sim \text{Po}(250)$. Since $250 > 15$, use normal approximation: $X \approx \text{N}(250, 250)$.
- Observed $x = 280$. With continuity correction: $P(X \geq 280) \approx P\left(Z > \frac{279.5-250}{\sqrt{250}}\right) = P(Z > 1.864)$.
- $P(Z > 1.864) \approx 0.0311$. Two-tailed p-value $= 2 \times 0.0311 = 0.0622$.
- Since $0.0622 > 0.01$, do not reject H_0. Insufficient evidence that the defect rate has changed.

EXERCISE 2 **Binomial and Poisson tests**

1. A dice is rolled 60 times and shows a six on 15 occasions. Test at the 5% level whether the dice is biased towards six.
2. A website receives hits at a mean rate of 12 per minute. In a 5-minute period, 75 hits are recorded. Test at the 5% level whether the rate has increased.
3. A manufacturer claims that 95% of his light bulbs last more than 1000 hours. A sample of 200 bulbs is tested and 185 last more than 1000 hours. Test at the 1% level whether the claim is exaggerated.

SOLUTION **Exercise 2**

1. $H_0 : p = 1/6, H_1 : p > 1/6$. $X \sim B(60, 1/6)$. $np_0 = 10 > 5$, normal approximation: $X \approx N(10, 8.333)$. $x = 15$, continuity correction: $z = \frac{14.5-10}{\sqrt{8.333}} = 1.559$. p-value $= P(Z > 1.559) = 0.0594 > 0.05$. Do not reject H_0. Insufficient evidence of bias.
2. $H_0 : \lambda = 12, H_1 : \lambda > 12$. Over 5 min: $X \sim Po(60)$. $x = 75$. Normal approximation: $X \approx N(60, 60)$. $z = \frac{74.5-60}{\sqrt{60}} = 1.873$. p-value $= P(Z > 1.873) = 0.0305 < 0.05$. Reject H_0. Sufficient evidence that rate has increased.
3. $H_0 : p = 0.95, H_1 : p < 0.95$. $X \sim B(200, 0.95)$. $np_0 = 190 > 5$. Normal approximation: $X \approx N(190, 9.5)$. $x = 185$, continuity correction: $z = \frac{185.5-190}{\sqrt{9.5}} = -1.461$. p-value $= P(Z < -1.461) = 0.0720 > 0.01$. Do not reject H_0. Insufficient evidence that claim is exaggerated.

3 TESTING A POPULATION MEAN

3.1 Case 1: Normal Population, Known Variance σ^2

When the population is normally distributed with known variance σ^2, the sample mean $\bar{X}$ is exactly normally distributed.

PROPERTY **Z-test for a population mean (σ^2 known)**

For a random sample $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$ with σ^2 known, testing $H_0 : \mu = \mu_0$:

- Test statistic: $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$ under H_0.
- Critical values: $\pm z_{\alpha/2}$ (two-tailed), $-z_\alpha$ (lower), $+z_\alpha$ (upper).
- p-value: from standard normal distribution.

EXEMPLE **Z-test: known variance, two-tailed**

A machine fills bags with cement. The weight is normally distributed with standard deviation $\sigma = 0.5$ kg. The target mean is 50 kg. A sample of 25 bags has mean $\bar{x} = 49.7$ kg. Test at the 5% level whether the mean differs from 50 kg.

- $H_0 : \mu = 50, H_1 : \mu \neq 50, \alpha = 0.05$.

- $Z = \frac{49.7-50}{0.5/\sqrt{25}} = \frac{-0.3}{0.1} = -3.0$.
- Two-tailed critical values: ± 1.96. Since $-3.0 < -1.96$, reject H_0.
- p-value $= 2 \times P(Z < -3.0) = 2 \times 0.00135 = 0.0027 < 0.05$.
- Conclusion: Reject H_0 at 5% level. Sufficient evidence that the mean weight differs from 50 kg.

3.2 Case 2: Normal Population, Unknown Variance (Small Sample)

When the population is normal but σ^2 is unknown, we use the sample variance S^2 and the t-distribution.

PROPERTY **t-test for a population mean (σ^2 unknown, normal population)**

For a random sample from $N(\mu, \sigma^2)$ with σ^2 unknown, testing $H_0 : \mu = \mu_0$:

- Test statistic: $T = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$ under H_0, where $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$.
- Critical values: from t-tables with $n - 1$ degrees of freedom.
- This test is exact for any sample size n when the population is normal.

WARNING **Assumption of normality**

The t-test requires the population to be normally distributed. For non-normal populations, it is only approximate unless n is large (see Case 3).

WORKED EXAMPLE 4 **t-test: small sample, unknown variance**

The breaking strength of a new rope is normally distributed. A sample of 10 ropes has mean $\bar{x} = 1850$ N and standard deviation $s = 120$ N. Test at the 5% level whether the mean breaking strength exceeds 1800 N.

SOLUTION

Solution.

- $H_0 : \mu = 1800$, $H_1 : \mu > 1800$ (upper one-tailed), $\alpha = 0.05$.
- $T = \frac{1850-1800}{120/\sqrt{10}} = \frac{50}{37.95} = 1.317$.

- Degrees of freedom = 9. Upper 5% critical value $t_{9,0.05} = 1.833$.
- Since $1.317 < 1.833$, do not reject H_0.
- p-value = $P(T_9 > 1.317) \approx 0.107 > 0.05$.
- Conclusion: Insufficient evidence that the mean breaking strength exceeds 1800 N.

3.3 Case 3: Large Sample, Any Population (Central Limit Theorem)

When n is large ($n \geq 30$ is the usual rule), the Central Limit Theorem guarantees that $\bar{X}$ is approximately normal regardless of the population distribution. If σ^2 is unknown, we estimate it by S^2.

PROPERTY **Large-sample Z-test for a population mean**

For a random sample of size $n \geq 30$ from any population with mean μ and variance σ^2, testing $H_0 : \mu = \mu_0$:

- Test statistic: $Z = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}} \approx N(0,1)$ under H_0.
- This is an approximate test; the approximation improves as n increases.
- Critical values and p-values from standard normal distribution.

EXEMPLE **Large-sample test: non-normal population**

The time (in minutes) customers spend in a bank is highly skewed. A sample of 50 customers has mean $\bar{x} = 12.4$ and standard deviation $s = 4.2$. Test at the 5% level whether the mean time exceeds 11 minutes.

- $H_0 : \mu = 11$, $H_1 : \mu > 11$, $\alpha = 0.05$.
- $n = 50 \geq 30 \Rightarrow$ CLT applies. $Z = \dfrac{12.4-11}{4.2/\sqrt{50}} = \dfrac{1.4}{0.594} = 2.357$.
- Upper 5% critical value = 1.645. Since $2.357 > 1.645$, reject H_0.
- p-value = $P(Z > 2.357) \approx 0.0092 < 0.05$.
- Conclusion: Reject H_0 at 5% level. Sufficient evidence that the mean time exceeds 11 minutes.

KEY POINTS **Choosing the correct test for a single mean**

Situation	Test statistic	Distribution under H_0
Normal population, σ^2 known	$Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$	N(0, 1) (exact)
Normal population, σ^2 unknown, any n	$T = \frac{\bar{X}-\mu_0}{S/\sqrt{n}}$	t_{n-1} (exact)
Non-normal population, $n \geq 30$	$Z = \frac{\bar{X}-\mu_0}{S/\sqrt{n}}$	N(0, 1) (approx., by CLT)

EXERCISE 3 **Tests for a population mean**

1. The lifetime of a battery is normally distributed with $\sigma = 2.5$ hours. A sample of 16 batteries has mean 48.2 hours. Test at the 5% level whether the mean lifetime is less than 50 hours.

2. A random sample of 12 students from a large university yields a mean height of 168.5 cm with standard deviation 6.2 cm. Assuming heights are normal, test at the 1% level whether the mean height differs from 165 cm.

3. The waiting time at a clinic is not normally distributed. A sample of 40 patients has mean 22.3 min and standard deviation 8.1 min. Test at the 5% level whether the mean waiting time is different from 20 min.

SOLUTION **Exercise 3**

1. $H_0 : \mu = 50$, $H_1 : \mu < 50$. $Z = \frac{48.2-50}{2.5/\sqrt{16}} = -2.88$. Critical value -1.645. $-2.88 < -1.645 \Rightarrow$ reject H_0. p-value $= 0.0020$. Sufficient evidence mean < 50 hours.

2. $H_0 : \mu = 165$, $H_1 : \mu \neq 165$. $T = \frac{168.5-165}{6.2/\sqrt{12}} = 1.952$. $df = 11$, two-tailed 1% critical values ± 3.106. 1.952 not in critical region. p-value $\approx 0.076 > 0.01$. Do not reject H_0. Insufficient evidence mean $\neq 165$ cm.

3. $H_0 : \mu = 20$, $H_1 : \mu \neq 20$. $n = 40 \geq 30$, CLT. $Z = \frac{22.3-20}{8.1/\sqrt{40}} = 1.795$. Two-tailed 5% critical ± 1.96. $1.795 < 1.96 \Rightarrow$ do not reject. p-value $= 2 \times 0.0363 = 0.0726 > 0.05$. Insufficient evidence mean $\neq 20$ min.

4 TESTING THE DIFFERENCE BETWEEN TWO MEANS (LARGE SAMPLES)

4.1 Two Independent Samples

We now compare the means of two independent populations. Let population 1 have mean μ_1 and variance σ_1^2, population 2 have mean μ_2 and variance σ_2^2. We take independent random samples of sizes n_1 and n_2, obtaining sample means $\bar{X}_1, \bar{X}_2$ and sample variances S_1^2, S_2^2.

PROPERTY **Sampling distribution of $\bar{X}_1 - \bar{X}_2$**

If the samples are independent and either (a) both populations are normal, or (b) both sample sizes are large ($n_1, n_2 \geq 30$), then

$$\bar{X}_1 - \bar{X}_2 \sim \mathrm{N}\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right) \quad \text{(exactly or approximately).}$$

PROPERTY **Two-sample Z-test for difference of means (large samples)**

For independent large samples ($n_1, n_2 \geq 30$) from any populations, testing $H_0 : \mu_1 - \mu_2 = \delta_0$ (usually $\delta_0 = 0$):

- Test statistic: $Z = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \approx \mathrm{N}(0,1)$ under H_0.
- If population variances σ_1^2, σ_2^2 are known, use them instead of S_1^2, S_2^2.
- Critical values and p-values from standard normal distribution.

EXEMPLE **Two-sample test: known variances**

Two brands of fertilizer are tested. Brand A: $\sigma_A = 1.2$ kg, $n_A = 40$, $\bar{x}_A = 25.4$ kg. Brand B: $\sigma_B = 1.5$ kg, $n_B = 35$, $\bar{x}_B = 23.8$ kg. Test at 5% whether Brand A gives a higher mean yield.

- $H_0 : \mu_A - \mu_B = 0$, $H_1 : \mu_A - \mu_B > 0$, $\alpha = 0.05$.
- $Z = \dfrac{(25.4-23.8)-0}{\sqrt{1.2^2/40 + 1.5^2/35}} = \dfrac{1.6}{\sqrt{0.036+0.0643}} = \dfrac{1.6}{0.3167} = 5.05$.
- Critical value 1.645. $5.05 > 1.645 \Rightarrow$ reject H_0.
- p-value $\approx 2.2 \times 10^{-7}$. Very strong evidence Brand A is better.

WORKED EXAMPLE 5 **Two-sample test: unknown variances, large samples**

A study compares the mean monthly expenditure on mobile data in two cities. Douala: $n_1 = 60$, $\bar{x}_1 = 4500$ FCFA, $s_1 = 1200$ FCFA. Yaoundé: $n_2 = 55$, $\bar{x}_2 = 4200$ FCFA, $s_2 = 1100$ FCFA. Test at the 5% level whether the mean expenditures differ.

SOLUTION

Solution.

- $H_0 : \mu_1 - \mu_2 = 0$, $H_1 : \mu_1 - \mu_2 \neq 0$ (two-tailed), $\alpha = 0.05$.
- Both $n_1, n_2 \geq 30 \Rightarrow$ CLT applies. Use sample variances.
- Standard error: $\sqrt{\frac{1200^2}{60} + \frac{1100^2}{55}} = \sqrt{24000 + 22000} = \sqrt{46000} = 214.48$.
- $Z = \frac{(4500-4200)-0}{214.48} = \frac{300}{214.48} = 1.399$.
- Two-tailed critical values ± 1.96. Since $|1.399| < 1.96$, do not reject H_0.
- p-value $= 2 \times P(Z > 1.399) = 2 \times 0.081 = 0.162 > 0.05$.
- Conclusion: Insufficient evidence that the mean monthly expenditures differ between the two cities.

4.2 Two-Sample t-Test (Small Samples, Normal Populations, Equal Variances)

Note: The official syllabus specifies "large scale (Using Central Limit theorem)" for the difference of two means. The t-test for small samples with equal variances is included here for completeness but the examinable focus is the large-sample Z-test.

PROPERTY **Two-sample t-test (equal variances, normal populations)**

If both populations are normal with equal variances $\sigma_1^2 = \sigma_2^2 = \sigma^2$, and samples are small, we pool the variances:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}, \qquad T = \frac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2} \text{ under } H_0.$$

WARNING **Equal variance assumption**

The pooled t-test assumes $\sigma_1^2 = \sigma_2^2$. If this is doubtful and samples are small, use Welch's approximate t-test (not in syllabus). For large samples, the Z-test with separate variances is robust and preferred.

4.3 Paired Samples (Matched Pairs)

When observations come in natural pairs (before/after, twins, matched subjects), we analyse the differences $D_i = X_{1i} - X_{2i}$. This reduces to a one-sample test on the mean difference μ_D.

PROPERTY **Paired t-test**

For n paired differences $D_1, \ldots, D_n$ from a normal distribution, testing $H_0 : \mu_D = 0$:

- Test statistic: $T = \dfrac{\bar{D}}{S_D/\sqrt{n}} \sim t_{n-1}$ under H_0.
- For large n, use $Z = \dfrac{\bar{D}}{S_D/\sqrt{n}} \approx \mathrm{N}(0,1)$.

EXEMPLE **Paired test: before and after**

A new teaching method is tried on 15 students. Their test scores before and after are recorded. The mean difference (after − before) is $\bar{d} = 4.2$ with $s_d = 5.8$. Test at 5% whether the method improves scores.

- $H_0 : \mu_D = 0$, $H_1 : \mu_D > 0$, $\alpha = 0.05$.
- $T = \dfrac{4.2}{5.8/\sqrt{15}} = \dfrac{4.2}{1.497} = 2.806$.
- $df = 14$, upper 5% critical value = 1.761. $2.806 > 1.761 \Rightarrow$ reject H_0.
- p-value $= P(T_{14} > 2.806) \approx 0.007 < 0.05$.
- Conclusion: Sufficient evidence that the teaching method improves scores.

KEY POINTS **Summary: Tests for difference of two means**

Situation	Test statistic	Distribution
Independent, large samples ($n_1, n_2 \geq 30$)	$Z = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$	$\mathrm{N}(0,1)$ (approx.)
Independent, normal, σ_1^2, σ_2^2 known	$Z = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$	$\mathrm{N}(0,1)$ (exact)
Independent, normal, equal variances, small n	$T = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{S_p\sqrt{1/n_1 + 1/n_2}}$	$t_{n_1+n_2-2}$ (exact)
Paired, normal differences	$T = \dfrac{\bar{D}}{S_D/\sqrt{n}}$	t_{n-1} (exact)
Paired, large n	$Z = \dfrac{\bar{D}}{S_D/\sqrt{n}}$	$\mathrm{N}(0,1)$ (approx.)

EXERCISE 4 Tests for difference of two means

1. Two types of tyres are tested for wear. Type A: $n_1 = 50$, $\bar{x}_1 = 42000$ km, $s_1 = 3500$ km. Type B: $n_2 = 45$, $\bar{x}_2 = 40500$ km, $s_2 = 3200$ km. Test at 5% whether Type A lasts longer.
2. A drug is tested on 20 patients. Systolic blood pressure is measured before and after. Mean reduction $\bar{d} = 8.5$ mmHg, $s_d = 6.2$ mmHg. Test at 1% whether the drug reduces blood pressure.
3. Two machines fill jars of jam. Machine 1: $\sigma_1 = 2.0$ g, $n_1 = 30$, $\bar{x}_1 = 502$ g. Machine 2: $\sigma_2 = 2.5$ g, $n_2 = 30$, $\bar{x}_2 = 498$ g. Test at 5% whether the machines have different mean fills.

SOLUTION Exercise 4

1. $H_0 : \mu_A - \mu_B = 0$, $H_1 : \mu_A - \mu_B > 0$. $Z = \dfrac{42000-40500}{\sqrt{3500^2/50+3200^2/45}} = \dfrac{1500}{\sqrt{245000+227556}} = \dfrac{1500}{687.5} = 2.182$. Critical 1.645. Reject H_0. p-value $= 0.0146$. Sufficient evidence Type A lasts longer.
2. $H_0 : \mu_D = 0$, $H_1 : \mu_D > 0$. $T = \dfrac{8.5}{6.2/\sqrt{20}} = 6.13$. $df = 19$, upper 1% critical $= 2.539$. Reject H_0. p-value < 0.001. Sufficient evidence drug reduces blood pressure.
3. $H_0 : \mu_1 - \mu_2 = 0$, $H_1 : \mu_1 - \mu_2 \neq 0$. $Z = \dfrac{502-498}{\sqrt{2^2/30+2.5^2/30}} = \dfrac{4}{\sqrt{0.1333+0.2083}} = \dfrac{4}{0.5846} = 6.84$. Critical ± 1.96. Reject H_0. p-value ≈ 0. Very strong evidence mean fills differ.

EXERCISE 5 Mixed revision exercises

1. A coin is tossed 100 times and lands heads 58 times. Test at the 5% level whether the coin is biased.
2. The number of accidents per month at a factory follows a Poisson distribution with mean 3. After a safety campaign, the next 6 months have totals: 2, 1, 0, 3, 1, 2. Test at 5% whether the accident rate has decreased.
3. A sample of 25 observations from a normal population with $\sigma = 4$ gives $\bar{x} = 52.3$. Test $H_0 : \mu = 50$ vs $H_1 : \mu > 50$ at 5%.
4. Two large samples give: $n_1 = 80$, $\bar{x}_1 = 15.2$, $s_1 = 3.1$; $n_2 = 75$, $\bar{x}_2 = 14.5$, $s_2 = 2.8$. Test $H_0 : \mu_1 = \mu_2$ vs $H_1 : \mu_1 \neq \mu_2$ at 5%.

SOLUTION **Exercise 5**

1. $H_0 : p = 0.5, H_1 : p \neq 0.5$. $X \sim B(100, 0.5) \approx N(50, 25)$. $z = \frac{57.5-50}{5} = 1.5$ (with continuity correction). Two-tailed p-value $= 2 \times 0.0668 = 0.1336 > 0.05$. Do not reject H_0. Insufficient evidence of bias.

2. $H_0 : \lambda = 3, H_1 : \lambda < 3$. Total over 6 months: $X = 9$. Under H_0, $X \sim Po(18)$. p-value $= P(X \leq 9) \approx 0.0154 < 0.05$. Reject H_0. Sufficient evidence accident rate has decreased.

3. $Z = \frac{52.3-50}{4/\sqrt{25}} = \frac{2.3}{0.8} = 2.875$. Critical 1.645. Reject H_0. p-value $= 0.0020$. Sufficient evidence $\mu > 50$.

4. $Z = \frac{15.2-14.5}{\sqrt{3.1^2/80+2.8^2/75}} = \frac{0.7}{\sqrt{0.1201+0.1045}} = \frac{0.7}{0.4738} = 1.477$. Critical ± 1.96. Do not reject H_0. p-value $= 0.140$. Insufficient evidence means differ.

5 TESTING A PROBABILITY AND THE MEAN OF A POISSON DISTRIBUTION

In the previous section we built the machinery of significance testing: the null hypothesis H_0, the alternative hypothesis H_1, the significance level α, the critical region, and the two types of error. We now put that machinery to work in the two simplest and most elegant situations of the syllabus: testing a claim about a **probability** p (using the **binomial distribution**) and testing a claim about the **mean of a Poisson distribution**. In both cases the test statistic has a known discrete distribution under H_0, so we can compute exact tail probabilities — no approximation is needed.

5.1 Testing a Probability: the Binomial Test

A concrete situation.

A factory in Douala claims that only 3% of the sachets it produces are defective. A consumer-protection inspector suspects the true proportion is higher. She takes a random sample of n sachets and counts X, the number of defective ones. If the claim is true, X should behave like a binomial random variable with parameters n and $p_0 = 0.03$. If the observed value of X is suspiciously large for such a distribution, the claim becomes untenable.

PROPERTY **Distribution of the test statistic under H_0**

Let X be the number of "successes" in n independent trials. To test

$$H_0 : p = p_0 \qquad \text{against} \qquad H_1 : p > p_0 \ (\text{or } p < p_0, \text{ or } p \neq p_0),$$

we use the fact that, under H_0,

$$X \sim B(n, p_0).$$

The **p-value** of an observed value x is a tail probability of this distribution:

- upper-tailed test ($H_1 : p > p_0$): p-value = $P(X \geq x)$;
- lower-tailed test ($H_1 : p < p_0$): p-value = $P(X \leq x)$;
- two-tailed test ($H_1 : p \neq p_0$): the p-value is $2 \times \min\big(P(X \geq x),\, P(X \leq x)\big)$ (equivalently, compare each tail with $\alpha/2$).

We reject H_0 at level α when the p-value is less than α. (The proof is simply the definition of the binomial model: independent trials, constant probability p_0.)

WARNING **The p-value is a tail probability, not a single probability**

A very common error is to compute $P(X = x)$ and compare it with α. This is wrong: even under H_0, any single value of a discrete variable can have a small probability. The correct question is: "If H_0 is true, how likely is a result as extreme or more extreme than the one observed?" That is exactly $P(X \geq x)$ (upper tail) or $P(X \leq x)$ (lower tail).

Finding the critical region.

Because X is discrete, we cannot usually hit α exactly. For an upper-tailed test at level α, the **critical value** is the **smallest integer k such that**

$$P(X \geq k) \leq \alpha, \qquad \text{i.e.} \qquad P(X \leq k-1) \geq 1-\alpha.$$

For a lower-tailed test, it is the **largest integer k such that** $P(X \leq k) \leq \alpha$. For a two-tailed test, place $\alpha/2$ in each tail. These cumulative probabilities are read from binomial tables or computed from $P(X = r) = \binom{n}{r} p_0^r (1-p_0)^{n-r}$.

METHOD **Binomial test of a probability**

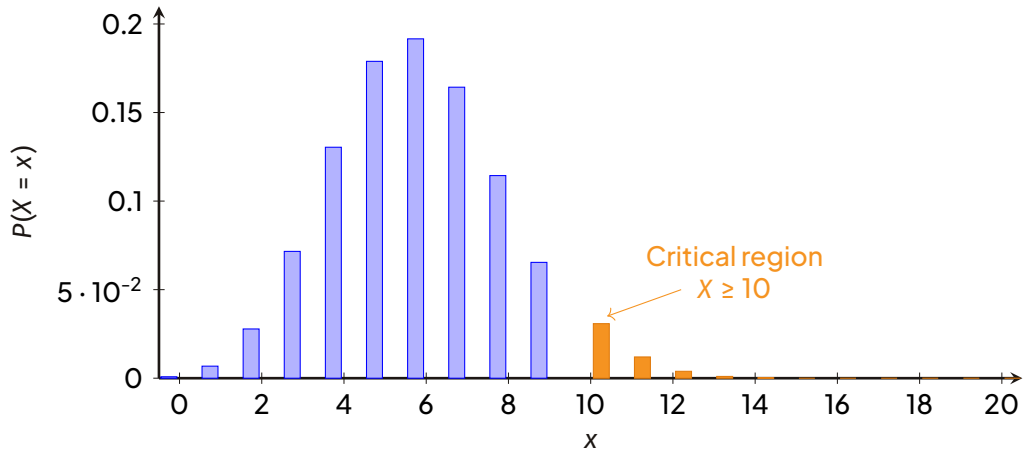
1. State the hypotheses: $H_0 : p = p_0$ and H_1 (one- or two-tailed, from the wording of the question). Define p clearly in context.
2. State the distribution under H_0: $X \sim B(n, p_0)$, and the significance level α.
3. Compute the p-value: the cumulative tail probability $P(X \geq x)$ or $P(X \leq x)$ for the observed x (double the smaller tail for a two-tailed test). Or determine the critical region and check whether x lies in it.
4. Decide: if p-value $< \alpha$ (or x in the critical region), reject H_0; otherwise do not reject H_0.
5. Conclude in context: one sentence, in the language of the problem, mentioning the strength of the evidence.

EXEMPLE **Critical region for $B(20, 0.3)$ at the 5% level**

A coin-like device is suspected of giving "success" with probability greater than 0.3. We test $H_0 : p = 0.3$ against $H_1 : p > 0.3$ using $n = 20$ trials at the 5% level. Under H_0, $X \sim B(20, 0.3)$. From cumulative binomial tables:

$$P(X \geq 9) = 0.1133 > 0.05, \qquad P(X \geq 10) = 0.0480 \leq 0.05.$$

The smallest k with $P(X \geq k) \leq 0.05$ is $k = 10$. The critical region is therefore $X \geq 10$, and the actual significance level of the test is 0.0480, not exactly 0.05.



Distribution $B(20, 0.3)$ under H_0. The orange bars form the upper-tail critical region $X \geq 10$ for a test at the 5% level (actual level 0.0480).

WORKED EXAMPLE 6 **Defective sachets in a Douala factory**

A factory in Douala claims that 3% of its sachets are defective. An inspector believes the true proportion is higher. In a random sample of 50 sachets, 4 are defective. Test at the 5% level of significance whether the claim should be rejected.

SOLUTION

Solution.
Step 1 — Hypotheses. Let p be the true proportion of defective sachets.

$$H_0 : p = 0.03 \qquad H_1 : p > 0.03.$$

Step 2 — Distribution under H_0. If X is the number of defective sachets in the sample, then under H_0, $X \sim B(50, 0.03)$; $\alpha = 0.05$.

Step 3 — p-value. The observed value is $x = 4$:

$$\begin{aligned} P(X \geq 4) &= 1 - P(X \leq 3) \\ &= 1 - \left[0.97^{50} + 50(0.03)(0.97^{49}) + \binom{50}{2}(0.03)^2(0.97^{48}) + \binom{50}{3}(0.03)^3(0.97^{47})\right] \\ &\approx 1 - (0.2181 + 0.3372 + 0.2555 + 0.1264) = 1 - 0.9372 = 0.0628. \end{aligned}$$

Step 4 — Decision. Since $0.0628 > 0.05$, we do not reject H_0.
Step 5 — Conclusion. At the 5% level, there is insufficient evidence that the proportion of defective sachets exceeds 3%; the inspector's suspicion is not confirmed by this sample.

WARNING **Two-tailed discrete tests: the actual level is rarely α**

Because the binomial distribution is discrete, tail probabilities jump in steps. The critical region "$P(X \geq k) \leq \alpha$" usually gives an actual significance level strictly smaller than α (e.g. 0.0480 instead of 0.05). In a two-tailed test, the two tails generally have different actual probabilities. When a GCE question asks for the significance level of a test whose critical region is given, compute the actual level by summing the tail probabilities — do not simply quote the nominal α.

5.2 Testing the Mean of a Poisson Distribution

A concrete situation.

The emergency unit of a hospital in Buea has, for years, received an average of 4 patients per day of a certain category. Staff now feel the rate has increased. Arrivals of this kind are well modelled by a Poisson distribution: they occur randomly, independently, and at a constant average rate. If X is the number of arrivals on a randomly chosen day, then under the "old" regime $X \sim Po(4)$. A day (or a period) with an unusually large count casts doubt on $\lambda = 4$.

PROPERTY **Poisson test of a mean**

Let X count events in a fixed interval, modelled by a Poisson distribution. To test

$$H_0 : \lambda = \lambda_0 \qquad \text{against} \qquad H_1 : \lambda > \lambda_0 \ (\text{or } \lambda < \lambda_0, \text{ or } \lambda \neq \lambda_0),$$

we use the fact that, under H_0, $X \sim Po(\lambda_0)$, and the p-value of an observed count x is a tail probability:

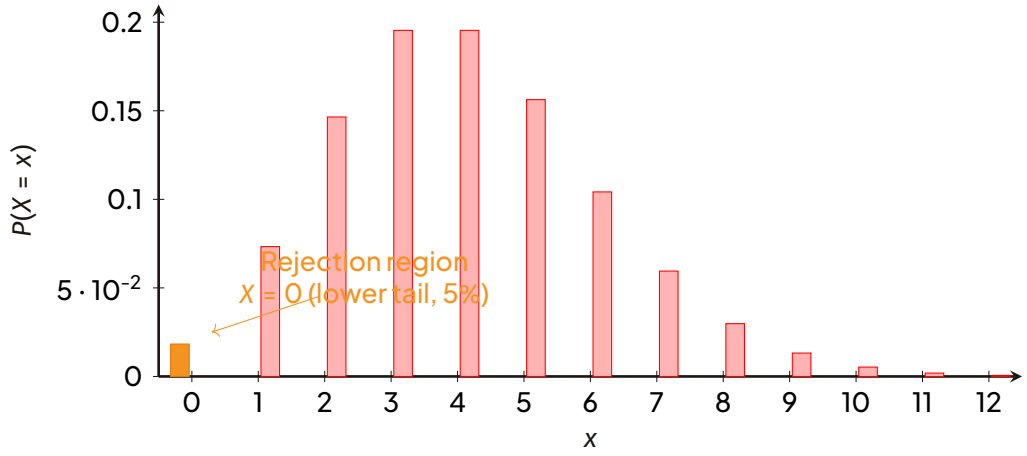
- upper tail ($H_1 : \lambda > \lambda_0$): p-value $= P(X \geq x)$;
- lower tail ($H_1 : \lambda < \lambda_0$): p-value $= P(X \leq x)$;
- two-tailed: the p-value is $2 \times \min\big(P(X \geq x),\ P(X \leq x)\big)$ (or compare each tail with $\alpha/2$).

If the observation covers t times the basic interval, use $X \sim Po(t\lambda_0)$ under H_0 (additive

property of the Poisson distribution).

METHOD **Poisson test of a mean**

1. State the hypotheses $H_0 : \lambda = \lambda_0$, H_1 as appropriate, defining λ with its unit interval ("mean number of arrivals per day").
2. State the model under H_0: $X \sim Po(\lambda_0)$ (adjusting for the length of the observation period if necessary), and give α.
3. Compute the p-value using $P(X = r) = e^{-\lambda_0}\dfrac{\lambda_0^r}{r!}$ and cumulative Poisson tables; or find the critical region: smallest k with $P(X \geq k) \leq \alpha$ (upper tail) or largest k with $P(X \leq k) \leq \alpha$ (lower tail).
4. Decide by comparing the p-value with α.
5. Conclude in context.



Distribution $Po(4)$ under H_0. Since $P(X \leq 0) = 0.0183 \leq 0.05$ but $P(X \leq 1) = 0.0916 > 0.05$, the lower-tail critical region at the 5% level is just $X = 0$.

WORKED EXAMPLE 7 **Arrivals at a Buea emergency unit**

The emergency unit of a hospital in Buea used to receive a mean of 4 cases per day of a certain category, modelled by a Poisson distribution. Management suspects the mean has increased. On a randomly chosen day, 9 such cases arrive. Test at the 5% level

whether the mean has risen above 4.

SOLUTION

Solution.
Step 1 — Hypotheses. Let λ be the mean number of such cases per day.

$$H_0 : \lambda = 4 \qquad H_1 : \lambda > 4.$$

Step 2 — Model under H_0. $X \sim Po(4)$, $\alpha = 0.05$.
Step 3 — p-value. From cumulative Poisson tables with $\lambda_0 = 4$:

$$P(X \geq 9) = 1 - P(X \leq 8) = 1 - 0.9786 = 0.0214.$$

Step 4 — Decision. Since $0.0214 < 0.05$, we reject H_0.
Step 5 — Conclusion. At the 5% level, there is significant evidence that the mean daily number of arrivals has increased above 4; the hospital should review its staffing accordingly.
Remark. Note that $P(X \geq 8) = 0.0511 > 0.05$: had only 8 cases arrived, the evidence would not have been significant. The critical region for this test is $X \geq 9$, with actual level 0.0214.

EXERCISE 6 **A two-tailed Poisson test**

Calls arrive at a call centre in Yaoundé at a mean rate of 3 per minute, modelled by a Poisson distribution. After a system change, the manager wants to test, at the 10% level, whether the rate has changed (in either direction). In a given minute, 7 calls arrive. Given $P(X \leq 6) = 0.9665$ and $P(X \leq 7) = 0.9881$ for $Po(3)$, carry out the test.

SOLUTION **Exercise 1**

$H_0 : \lambda = 3$; $H_1 : \lambda \neq 3$ (two-tailed, 5% in each tail). Upper tail: $P(X \geq 7) = 1 - P(X \leq 6) = 1 - 0.9665 = 0.0335$. Since $0.0335 < 0.05$, the observed value falls in the upper rejection region (equivalently, doubled p-value $0.067 < 0.10$). We reject H_0: at the 10% level there is evidence that the call rate has changed — indeed increased.

5.3 Large Samples: the Normal Approximation (Extension)

Exact binomial and Poisson computations become heavy when n or λ is large. The following approximations, already met in the distribution chapters, are exam-useful:

KEY POINTS **Normal approximations with continuity correction**

- If $X \sim B(n, p)$ with n large ($np > 5$ and $n(1-p) > 5$), then $X \approx N(np,\ np(1-p))$.
- If $X \sim Po(\lambda)$ with λ large (say $\lambda > 10$), then $X \approx N(\lambda,\ \lambda)$.

Because we approximate a discrete variable by a continuous one, apply a **continuity correction**:

$$P(X \geq x) \approx P\left(Z \geq \frac{x - 0.5 - \mu}{\sigma}\right), \qquad P(X \leq x) \approx P\left(Z \leq \frac{x + 0.5 - \mu}{\sigma}\right).$$

For instance, in the Douala example with $n = 500$ instead of 50, the exact binomial sum would be unbearable by hand, but $B(500, 0.03) \approx N(15,\ 14.55)$ gives the p-value in a few lines. Unless the question says otherwise at this level, however, prefer the exact cumulative probabilities whenever they are computable or given in tables.

DID YOU KNOW? **Why the GCE loves discrete tests**

Binomial and Poisson tests are favourites of examiners precisely because everything is exact: no estimated variance, no approximation. The whole logic of hypothesis testing — hypotheses, tail probabilities, critical regions, actual versus nominal significance levels — is visible in its purest form. Master these two tests and the tests on means in the next sections will feel like old friends wearing new clothes.

6 TESTING A POPULATION MEAN

In the previous section we tested hypotheses about a probability p and about the mean of a Poisson distribution, using exact discrete distributions (binomial and Poisson) to locate critical regions. We now turn to the most common situation in practice: testing a claim about the **mean** μ of a population. A water company in Douala claims its sachets contain on average 500 ml; a miller claims its bags of flour average 50 kg; a school claims its candidates' mean mark is 12/20. In each case the natural evidence is the sample mean $\bar{X}$, and the question is whether the observed $\bar{x}$ is compatible with the claimed value μ_0, or so far from it that the claim should be rejected.

6.1 Setting Up the Hypotheses

As before, the **null hypothesis** H_0 states the status quo — the claimed or standard value — while the **alternative hypothesis** H_1 expresses the suspicion we wish to test:

- two-tailed: $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ (the mean may have changed in either direction);
- one-tailed (right): $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$ (suspected increase);
- one-tailed (left): $H_0 : \mu = \mu_0$ against $H_1 : \mu < \mu_0$ (suspected decrease).

The test statistic is the sample mean $\bar{X}$ of a random sample $X_1, \ldots, X_n$. Small values of $|\bar{x} - \mu_0|$ support H_0; large values support H_1. To judge "large" we need the distribution of $\bar{X}$ when H_0 is true.

WARNING **Choosing the tail before seeing the data**

The direction of H_1 must be decided from the wording of the problem, before (or independently of) looking at the sample result. "Suspects the mean has decreased" gives a left-tailed test; "suspects the mean has changed" gives a two-tailed test. Choosing the tail after seeing that $\bar{x} < \mu_0$ in order to force a rejection is a serious methodological error — it roughly doubles the true significance level.

6.2 The Sampling Distribution of the Sample Mean

PROPERTY **Distribution of the sample mean**

If $X \sim N(\mu, \sigma^2)$ and $X_1, \ldots, X_n$ is a random sample, then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \qquad \text{so} \qquad Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

If the population is not normal but n is large (in practice $n \geq 30$), the Central Limit Theorem gives the same result approximately: $\bar{X} \overset{\text{approx}}{\sim} N(\mu, \sigma^2/n)$.

The proof of the exact case uses the fact that a linear combination of independent normal variables is normal, with $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$; the approximate case is precisely the Central Limit Theorem studied earlier. The key message is that **averaging reduces spread**: the standard deviation of $\bar{X}$, called the **standard error**, is $\sigma/\sqrt{n}$, so a large sample pins down μ much more sharply.

KEY POINTS **Which case am I in?**

Situation	Distribution of $\bar{X}$ under H_0	Standard error to use
Normal population, σ known	Exactly $N(\mu_0, \sigma^2/n)$	$\sigma/\sqrt{n}$
Any population, $n \geq 30$, σ known	Approximately $N(\mu_0, \sigma^2/n)$ (CLT)	$\sigma/\sqrt{n}$
Any population, $n \geq 30$, σ unknown	Approximately $N(\mu_0, \sigma^2/n)$ (CLT)	$s/\sqrt{n}$

In every case the test statistic is $z = \dfrac{\bar{x} - \mu_0}{\text{standard error}}$, compared with $N(0, 1)$.

6.3 Case 1: Normal Population, Known Variance — the z-Test

Suppose $X \sim N(\mu, \sigma^2)$ with σ^2 **known**, and we test $H_0 : \mu = \mu_0$. Under H_0,

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1).$$

We compute the observed value $z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ and compare it with critical values from standard normal tables, or compute the p-value.

KEY POINTS **Usual critical values of $N(0,1)$**

Significance level	10%	5%	2%	1%
One-tailed critical value	1.282	1.645	2.054	2.326
Two-tailed critical value	1.645	1.960	2.326	2.576

Reject H_0 when the observed z falls beyond the critical value in the direction(s) of H_1.

DEFINITION **p-value for a test on a mean**

The **p-value** is the probability, computed assuming H_0 true, of obtaining a value of the test statistic at least as extreme (in the direction of H_1) as the one observed. For a z-test with observed value z:

- right-tailed: $p = P(Z \geq z) = 1 - \Phi(z)$;
- left-tailed: $p = P(Z \leq z) = \Phi(z)$;
- two-tailed: $p = 2\,P(Z \geq |z|) = 2\big(1 - \Phi(|z|)\big)$.

We reject H_0 at level α exactly when $p \leq \alpha$.

WARNING **σ is not s**

Do not confuse the population standard deviation σ with the sample standard deviation s. Use σ only when it is genuinely known (given by the problem, or from long production experience). If σ is unknown and must be estimated from the sample, you are in Case 2 below — which is only justified for large n. Always state explicitly which quantity you are using and why.

WORKED EXAMPLE 8 **Known variance, two-tailed test**

The masses of bags of cement produced by a factory are normally distributed with standard deviation $\sigma = 0.9$ kg. The bags are labelled "50 kg". A random sample of 25 bags has mean mass 49.62 kg. Test, at the 5% level of significance, whether the mean mass differs from 50 kg.

SOLUTION

Solution. Step 1 — Hypotheses. "Differs from" means either direction, so the test is two-tailed:

$$H_0 : \mu = 50 \qquad \text{against} \qquad H_1 : \mu \neq 50, \qquad \alpha = 0.05.$$

Step 2 — Standard error. The population is normal and $\sigma = 0.9$ is known, so the standard error is $\sigma/\sqrt{n} = 0.9/\sqrt{25} = 0.18$ kg. Step 3 — Test statistic.

$$z = \frac{49.62 - 50}{0.18} = \frac{-0.38}{0.18} \approx -2.11.$$

Step 4 — Decision. The two-tailed critical values at 5% are ± 1.96. Since $|-2.11| = 2.11 > 1.96$, the observed value lies in the critical region. (The p-value is $2\big(1 - \Phi(2.11)\big) \approx 2(1 - 0.9826) = 0.0348 < 0.05$.) Step 5 — Conclusion. At the 5% level of significance, there is sufficient evidence that the mean mass of the bags differs from the labelled 50 kg; the machine should be checked and adjusted.

6.4 Case 2: Large Sample, Unknown Variance

In real life σ is rarely known. If the sample is large ($n \geq 30$), two facts save us:

1. by the Central Limit Theorem, $\bar{X}$ is approximately normal whatever the population shape;
2. for large n, the sample standard deviation s is a reliable estimate of σ, so replacing σ by s introduces negligible extra error.

The test statistic becomes

$$Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \overset{\text{approx}}{\sim} N(0,1) \quad \text{under } H_0,$$

and the test proceeds exactly as in Case 1.

METHOD **The z-test for a population mean**

1. State the hypotheses: $H_0 : \mu = \mu_0$ and the appropriate H_1 (one- or two-tailed), and the significance level α.
2. Choose the standard error: $\sigma/\sqrt{n}$ if σ is known; $s/\sqrt{n}$ if σ is unknown and $n \geq 30$.

3. Compute $z = \dfrac{\bar{x} - \mu_0}{\text{standard error}}$.
4. Decide: either compare $|z|$ (or z, one-tailed) with the critical value, or compute the p-value and compare with α.
5. Conclude in context: state whether H_0 is rejected or not, at the stated level, in the language of the original problem.

WORKED EXAMPLE 9 **The sachet-water claim**

A water company in Bonabéri claims that its sachets contain on average 500 ml. A consumers' association suspects the true mean is less than 500 ml. A random sample of 50 sachets gives $\bar{x} = 496.4$ ml with sample standard deviation $s = 9.8$ ml. Test the company's claim at the 5% level of significance.

SOLUTION

Solution. Step 1 — Hypotheses. The suspicion is a decrease, so the test is left-tailed:

$$H_0 : \mu = 500 \qquad \text{against} \qquad H_1 : \mu < 500, \qquad \alpha = 0.05.$$

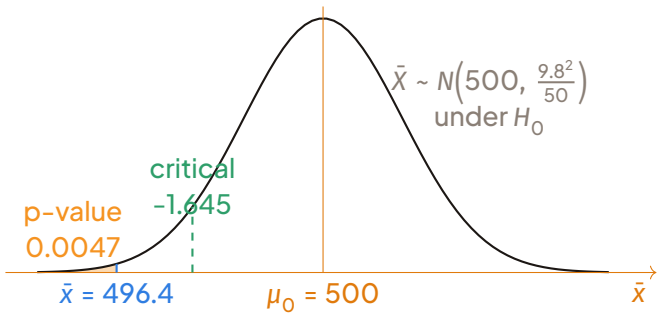
Step 2 — Standard error. σ is unknown but $n = 50 \geq 30$, so by the CLT we use $s/\sqrt{n} = 9.8/\sqrt{50} \approx 1.386$ ml. Step 3 — Test statistic.

$$z = \frac{496.4 - 500}{9.8/\sqrt{50}} = \frac{-3.6}{1.386} \approx -2.60.$$

Step 4 — Decision. The left-tailed critical value at 5% is -1.645. Since $-2.60 < -1.645$, the observed value lies in the critical region. Equivalently, the p-value is

$$p = P(Z \leq -2.60) = \Phi(-2.60) = 1 - \Phi(2.60) \approx 1 - 0.9953 = 0.0047 < 0.05.$$

Step 5 — Conclusion. We reject H_0 at the 5% level: there is significant evidence that the mean content of the sachets is less than the advertised 500 ml. The consumers' suspicion is supported.



Sampling distribution of $\bar{X}$ under H_0 for the sachet-water test. The observed $\bar{x} = 496.4$ lies beyond the critical value; the shaded area is the one-tailed p-value.

WORKED EXAMPLE 10 **A case where H_0 is not rejected**

A school states that the mean mark of its candidates in a mock examination is 12 out of 20. A random sample of 40 scripts gives $\bar{x} = 11.5$ with sample standard deviation $s = 3.2$. Test, at the 5% level, whether the mean mark differs from 12.

SOLUTION

Solution. Step 1 — Hypotheses. Two-tailed test:

$$H_0 : \mu = 12 \qquad \text{against} \qquad H_1 : \mu \neq 12, \qquad \alpha = 0.05.$$

Step 2 — Standard error. σ unknown, $n = 40 \geq 30$: standard error $= s/\sqrt{n} = 3.2/\sqrt{40} \approx 0.506$. Step 3 — Test statistic.

$$z = \frac{11.5 - 12}{0.506} \approx -0.99.$$

Step 4 — Decision. Two-tailed critical values at 5%: ± 1.96. Since $|-0.99| = 0.99 < 1.96$, the observed value does not lie in the critical region. (The p-value is $2\big(1 - \Phi(0.99)\big) \approx 2(1 - 0.8389) = 0.322 > 0.05$.) Step 5 — Conclusion. At the 5% level of significance, there is insufficient evidence against the school's claim: the sample is consistent with a mean mark of 12 out of 20. Note carefully: we have not proved that $\mu = 12$; the data simply do not contradict it.

6.5 The Link with Confidence Intervals

There is a beautiful and examinable connection between two-tailed tests and confidence intervals. A 95% confidence interval for μ (large sample) is

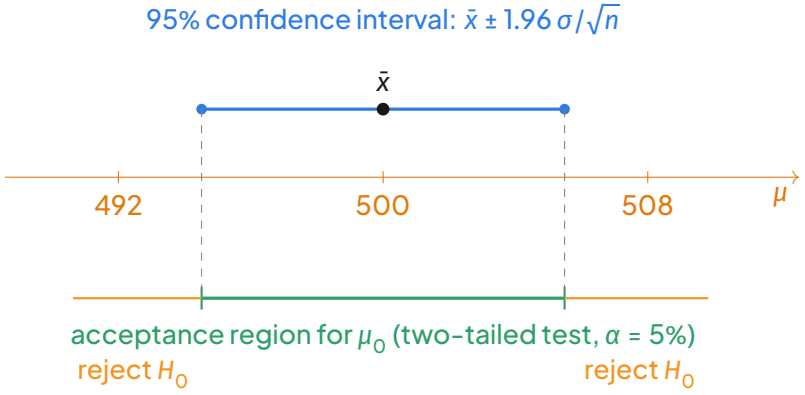
$$\bar{x} \pm 1.96\,\frac{\sigma}{\sqrt{n}}.$$

Now, the two-tailed test of $H_0 : \mu = \mu_0$ at the 5% level accepts H_0 exactly when

$$\left|\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}\right| < 1.96 \iff \mu_0 \in \left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right).$$

PROPERTY **Duality between tests and confidence intervals**

A two-tailed test of $H_0 : \mu = \mu_0$ at level α rejects H_0 exactly when μ_0 lies outside the $(1 - \alpha)$ confidence interval for μ. Equivalently, the confidence interval is precisely the set of values of μ_0 that would not be rejected.



The 95% confidence interval for μ and the acceptance region of the two-tailed 5% test have exactly the same endpoints: $H_0 : \mu = \mu_0$ is rejected precisely when μ_0 falls outside the interval.

For the sachet-water data, a 95% confidence interval for μ is $496.4 \pm 1.96 \times 1.386$, i.e. approximately $(493.7, 499.1)$ ml. The claimed value 500 lies outside this interval, so a two-tailed test at 5% would also reject H_0 — consistent with our one-tailed conclusion.

WARNING **Concluding properly**

A conclusion such as "reject H_0" alone earns little credit. The examiner expects three ingredients: (i) the decision (reject / do not reject H_0), (ii) the significance level used, and (iii) the context. Write, for example: "At the 5% level of significance, there is sufficient evidence to reject the company's claim that the mean content is 500 ml." Also remember: failing to reject H_0 is not proof that H_0 is true — the data are merely consistent with it.

DID YOU KNOW? **Small samples: the t-test**

When the population is normal but σ is unknown and the sample is small ($n < 30$), replacing σ by s adds too much uncertainty for the normal approximation. The correct statistic then follows Student's t-distribution with $n - 1$ degrees of freedom, introduced by W. S. Gosset (writing as "Student") while working for the Guinness brewery. The t-test is beyond this syllabus, but you may meet it at university — its logic (hypotheses, critical values, p-values, conclusion in context) is exactly the one you have mastered here.

EXERCISE 7

A factory produces rods whose lengths are normally distributed with standard deviation $\sigma = 1.2$ cm. The target mean length is 75 cm. A random sample of 36 rods has mean length 74.5 cm.

1. Test, at the 5% level, whether the mean length differs from 75 cm.
2. Compute the p-value of the test.
3. Give a 95% confidence interval for the mean length and verify the duality with part 1.

SOLUTION **Exercise 1**

1. $H_0 : \mu = 75$, $H_1 : \mu \neq 75$ (two-tailed). σ is known, so $z = \dfrac{74.5 - 75}{1.2/\sqrt{36}} = \dfrac{-0.5}{0.2} = -2.5$. Since $|-2.5| = 2.5 > 1.96$, reject H_0 at the 5% level: the mean length differs significantly from 75 cm. 2. $p = 2\big(1 - \Phi(2.5)\big) = 2(1 - 0.9938) = 0.0124 < 0.05$, confirming rejection. 3. CI: $74.5 \pm 1.96 \times 0.2 = (74.108,\ 74.892)$ cm. Since $75 \notin (74.108, 74.892)$, the two-tailed test rejects H_0 — exactly as found in part 1.

EXERCISE 8

A bakery claims that the mean mass of its loaves is 800 g. An inspector suspects the loaves are underweight. A random sample of 64 loaves gives a mean mass of 795 g with sample standard deviation 24 g.

1. State suitable null and alternative hypotheses.
2. Carry out the test at the 1% level of significance.
3. Would the conclusion change at the 5% level? Justify your answer.

SOLUTION **Exercise 2**

1. $H_0 : \mu = 800$ against $H_1 : \mu < 800$ (left-tailed, since underweight loaves are suspected). 2. σ unknown, $n = 64 \geq 30$: standard error $= 24/\sqrt{64} = 3$ g. Then $z = \dfrac{795 - 800}{3} \approx -1.67$. The left-tailed critical value at 1% is -2.326. Since $-1.67 > -2.326$, we do not reject H_0 at the 1% level: the evidence of underweight loaves is not strong enough at this level. 3. At 5% the critical value is -1.645, and $-1.67 < -1.645$, so H_0 would be rejected at the 5% level. The conclusion depends on the level chosen: the evidence is significant at 5% but not at the stricter 1% level. (The p-value is $\Phi(-1.67) \approx 0.0475$, which lies between 0.01 and 0.05, confirming both conclusions.)

7 TESTING THE DIFFERENCE BETWEEN TWO MEANS (LARGE SAMPLES)

In the previous section we learnt how to test a claim about one population mean: is the mean mass of a bag of cement really 50 kg? But in real life — and very often at the GCE — the interesting question is not about one population but about comparing two populations. Is the mean maize yield per hectare the same in Mezam Division as in Bui Division? Do customers at Bank A wait longer on average than customers at Bank B in Yaoundé? Is a new teaching method more effective than the old one?

In each case we take a sample from each population, compute the two sample means $\bar{x}_1$ and $\bar{x}_2$, and ask: is the observed difference $\bar{x}_1 - \bar{x}_2$ large enough to conclude that the population means μ_1 and μ_2 really differ, or could it be due to sampling fluctuation alone? This section answers that question rigorously, using the Central Limit Theorem.

7.1 Setting Up the Hypotheses

EXPERIMENT **Two regions, one question**

An agricultural officer wants to know whether the mean maize yield per hectare is the same in two regions of the North-West. She cannot measure every farm, so she samples $n_1 = 60$ farms in Region 1 and $n_2 = 80$ farms in Region 2, obtaining sample means $\bar{x}_1$ and $\bar{x}_2$ tonnes per hectare. Even if the two regions were identical, $\bar{x}_1$ and $\bar{x}_2$ would almost never be exactly equal, because each sample is only a fragment of its population. The whole art of the test is to decide whether the gap $\bar{x}_1 - \bar{x}_2$ is surprisingly large under the assumption that $\mu_1 = \mu_2$.

DEFINITION **Hypotheses for comparing two means**

Let μ_1 and μ_2 be the means of two populations. The null hypothesis is usually

$$H_0 : \mu_1 - \mu_2 = d_0 \qquad \text{(most often } d_0 = 0, \text{ i.e. } \mu_1 = \mu_2)$$

against one of the alternatives:

- $H_1 : \mu_1 - \mu_2 \neq d_0$ (two-tailed test);
- $H_1 : \mu_1 - \mu_2 > d_0$ (one-tailed test, right tail);
- $H_1 : \mu_1 - \mu_2 < d_0$ (one-tailed test, left tail).

The **test statistic** is the difference of the sample means, $\bar{X}_1 - \bar{X}_2$.

WARNING **Choosing the tail from the wording**

As for every significance test, the alternative hypothesis reflects the claim being investigated, decided before looking at the data:

- "differ", "is there a difference", "changed" ⇒ two-tailed;
- "greater than", "longer", "higher", "better" ⇒ right tail;
- "less than", "shorter", "lower" ⇒ left tail.

Never let the sign of the observed difference $\bar{x}_1 - \bar{x}_2$ dictate the choice of tail after the fact.

7.2 The Sampling Distribution of $\bar{X}_1 - \bar{X}_2$

To build a test we need the distribution of $\bar{X}_1 - \bar{X}_2$ when H_0 is true. Two facts from earlier chapters do all the work.

Expectation. By linearity of expectation (no independence needed):

$$E(\bar{X}_1 - \bar{X}_2) = E(\bar{X}_1) - E(\bar{X}_2) = \mu_1 - \mu_2.$$

Variance. Since $\mathrm{Var}(\bar{X}_i) = \sigma_i^2 / n_i$, and since the two samples are drawn independently, $\bar{X}_1$ and $\bar{X}_2$ are independent random variables. For independent variables, the variance of a difference is the sum of the variances:

$$\mathrm{Var}(\bar{X}_1 - \bar{X}_2) = \mathrm{Var}(\bar{X}_1) + \mathrm{Var}(\bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

PROPERTY **Sampling distribution of the difference of two means**

Let $\bar{X}_1$ and $\bar{X}_2$ be the means of independent samples of sizes n_1 and n_2 taken from popula-

tions with means μ_1, μ_2 and variances σ_1^2, σ_2^2. Then

$$E(\bar{X}_1 - \bar{X}_2) = \mu_1 - \mu_2, \qquad \operatorname{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

Moreover, by the **Central Limit Theorem**, when n_1 and n_2 are both large (in practice $n_1, n_2 \geq 30$),

$$\bar{X}_1 - \bar{X}_2 \approx N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

(The derivation of the expectation and variance formulae above is instructive and may be asked; the Central Limit Theorem itself is quoted.)

WARNING **The standard error of a difference**

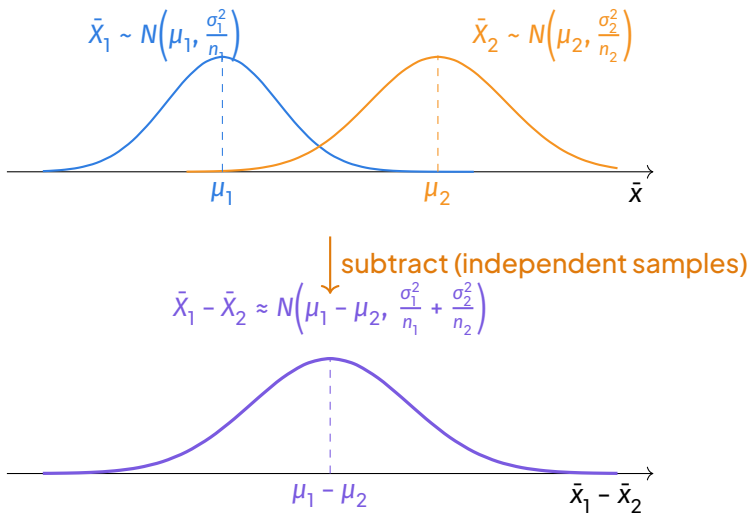
The standard error of $\bar{X}_1 - \bar{X}_2$ is

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, \qquad \text{not} \qquad \frac{\sigma_1}{\sqrt{n_1}} + \frac{\sigma_2}{\sqrt{n_2}}.$$

Variances add; standard deviations do not. A very common GCE error is to add the two standard errors of the means. Always add the variances first, then take the square root of the sum.

WARNING **Independence is essential**

The formula $\operatorname{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$ requires the two samples to be **independent**: different, unrelated individuals in each sample. If the same subjects are measured twice (for example the same students before and after a training programme), the data are paired, the two samples are dependent, and a different approach is required — this lies beyond the present syllabus.



Two independent sample means (top) and the sampling distribution of their difference (bottom): the difference is centred on $\mu_1 - \mu_2$ and its variance is the sum of the two variances.

7.3 The Two-Sample z-Test (Large Samples)

Under $H_0 : \mu_1 - \mu_2 = d_0$, the standardised statistic

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$$

follows approximately the standard normal distribution $N(0,1)$ when both samples are large. An observed value of Z falling in the critical region leads us to reject H_0.

METHOD **Two-sample z-test for the difference of means (large samples)**

1. State the hypotheses. $H_0 : \mu_1 - \mu_2 = d_0$ against the appropriate H_1 (one- or two-tailed). State the significance level α (usually 5% or 1%).
2. Check the conditions. The two samples are independent; n_1 and n_2 are both large (≥ 30).
3. Compute the test statistic.
$$z = \frac{\bar{x}_1 - \bar{x}_2 - d_0}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}.$$
If σ_1^2, σ_2^2 are unknown (the usual situation), replace them by the sample variances s_1^2, s_2^2; this is justified because the samples are large.

4. Find the critical region or the p-value. For a two-tailed test: reject H_0 if $|z| > 1.960$ at the 5% level, or $|z| > 2.576$ at the 1% level. For a one-tailed test: compare z with 1.645 (right tail) or -1.645 (left tail) at 5%, and with 2.326 or -2.326 at 1%.
5. Conclude in context. "Since $|z| = \ldots > 1.960$, we reject H_0 at the 5% level: there is significant evidence that the mean …differs between …and ….."

WORKED EXAMPLE 11 **Comparing maize yields in two regions**

An agricultural officer samples 60 farms in Region 1 and 80 farms in Region 2 of the North-West. The maize yields (tonnes per hectare) give

$$\bar{x}_1 = 2.85, \quad s_1 = 0.60 \qquad \text{and} \qquad \bar{x}_2 = 2.62, \quad s_2 = 0.72.$$

Test at the 5% level whether the mean yields of the two regions differ.

SOLUTION

Solution. 1. Hypotheses. $H_0 : \mu_1 - \mu_2 = 0$ against $H_1 : \mu_1 - \mu_2 \neq 0$ (two-tailed), at $\alpha = 5\%$.
2. Conditions. Independent samples, $n_1 = 60 \geq 30$, $n_2 = 80 \geq 30$: the Central Limit Theorem applies. The population variances are unknown, so we use s_1^2 and s_2^2.
3. Test statistic.

$$z = \frac{2.85 - 2.62 - 0}{\sqrt{\dfrac{0.60^2}{60} + \dfrac{0.72^2}{80}}} = \frac{0.23}{\sqrt{0.006 + 0.00648}} = \frac{0.23}{\sqrt{0.01248}} = \frac{0.23}{0.1117} \approx 2.06.$$

4. Decision. Two-tailed at 5%: critical values ±1.960. Since $|z| = 2.06 > 1.960$, we reject H_0. (Equivalently, the p-value is $2\,P(Z > 2.06) \approx 2(0.0197) = 0.039 < 0.05$.)
5. Conclusion. At the 5% level there is significant evidence that the mean maize yields per hectare differ between the two regions, Region 1 appearing higher. Note that at the 1% level ($2.06 < 2.576$) we would not reject H_0 — the evidence is significant at 5% but not at 1%.

WORKED EXAMPLE 12 **Waiting times at two banks in Yaoundé**

A consumer association measures the waiting time (in minutes) of customers at two banks. For Bank A: $n_1 = 50$ customers, $\bar{x}_1 = 18.4$, $s_1 = 5.2$. For Bank B: $n_2 = 40$ customers, $\bar{x}_2 = 15.9$, $s_2 = 4.6$. Test at the 1% level the claim that customers wait longer on average at Bank A.

SOLUTION

Solution. 1. Hypotheses. The claim "longer at A" is one-sided: $H_0 : \mu_1 - \mu_2 = 0$ against

$H_1 : \mu_1 - \mu_2 > 0$, at $\alpha = 1\%$.
2. Conditions. Independent samples, both sizes large: the Central Limit Theorem is valid.
3. Test statistic.

$$z = \frac{18.4 - 15.9}{\sqrt{\frac{5.2^2}{50} + \frac{4.6^2}{40}}} = \frac{2.5}{\sqrt{0.5408 + 0.529}} = \frac{2.5}{\sqrt{1.0698}} = \frac{2.5}{1.0343} \approx 2.42.$$

4. Decision. One-tailed at 1%: critical value 2.326. Since $z = 2.42 > 2.326$, we reject H_0. (p-value $= P(Z > 2.42) \approx 0.0078 < 0.01$.)
5. Conclusion. At the 1% level there is significant evidence that the mean waiting time at Bank A exceeds that at Bank B.

WORKED EXAMPLE 13 **Testing a claimed difference $d_0 \neq 0$**

A distributor claims that the mean lifetime of Brand P batteries exceeds that of Brand Q batteries by at least 10 hours. Tests on independent random samples give: Brand P, $n_1 = 64$, $\bar{x}_1 = 118$ hours, $s_1 = 16$ hours; Brand Q, $n_2 = 49$, $\bar{x}_2 = 104$ hours, $s_2 = 14$ hours. Test the distributor's claim at the 5% level.

SOLUTION

Solution. 1. Hypotheses. The claim is $\mu_1 - \mu_2 \geq 10$; we place it in H_0 and test whether the data contradict it: $H_0 : \mu_1 - \mu_2 = 10$ against $H_1 : \mu_1 - \mu_2 < 10$ (left tail), at $\alpha = 5\%$.
2. Conditions. Independent samples, $n_1 = 64$ and $n_2 = 49$ both large.
3. Test statistic. Here $d_0 = 10$:

$$z = \frac{118 - 104 - 10}{\sqrt{\frac{16^2}{64} + \frac{14^2}{49}}} = \frac{4}{\sqrt{4 + 4}} = \frac{4}{2.828} \approx 1.41.$$

4. Decision. Left tail at 5%: reject H_0 if $z < -1.645$. Here $z = 1.41 > -1.645$, so we do not reject H_0.
5. Conclusion. At the 5% level the data are consistent with the distributor's claim: there is no significant evidence that the mean lifetime advantage of Brand P over Brand Q is less than 10 hours. Note carefully the role of $d_0 = 10$ in the numerator: forgetting it would give $z = \frac{14}{2.828} \approx 4.95$ and a completely different analysis.

WARNING **What if the samples are small?**

The z-test above rests on the Central Limit Theorem, so it needs both n_1 and n_2 large. If the samples are small and the populations are normal with a common unknown variance, the correct tool is the two-sample t-test with a pooled variance — this is beyond the present

syllabus, but you should be able to recognise from the wording that a small-sample situation calls for a different method.

WARNING **"Not rejecting" is not "proving"**

When $|z|$ falls below the critical value, the correct conclusion is: there is insufficient evidence, at the stated level, against H_0. We have not proved that $\mu_1 = \mu_2$; the observed difference is simply small enough to be explained by sampling fluctuation. Avoid writing "H_0 is true" in your GCE script.

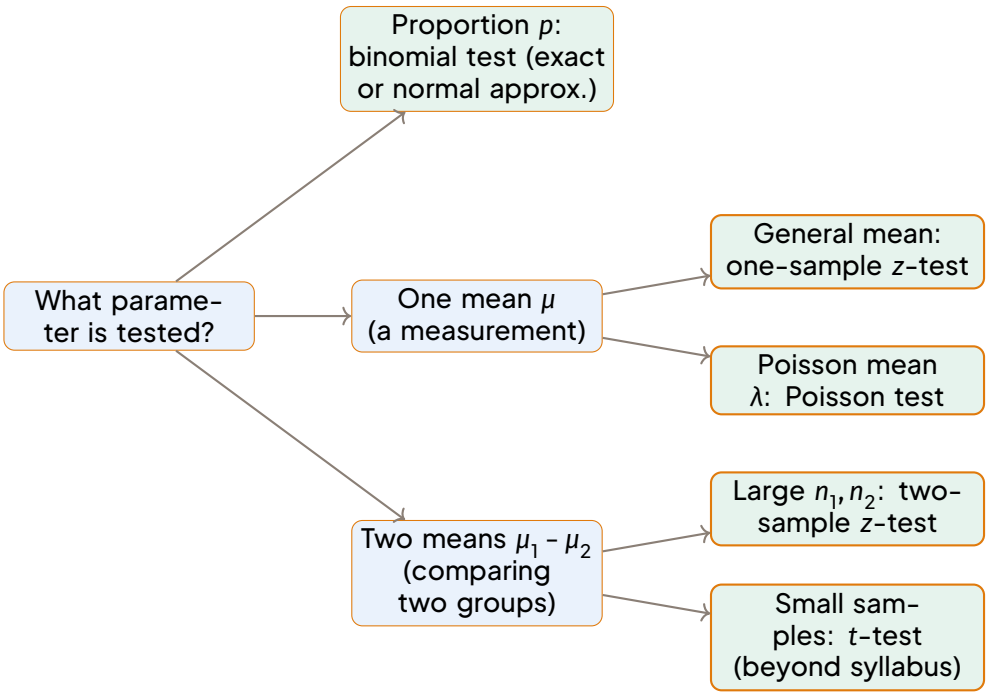
7.4 Chapter Synthesis: Choosing the Correct Test

A GCE hypothesis-testing question is half solved the moment you identify which parameter is being tested and which distribution models the situation. Read the wording and ask yourself: What is the parameter? What is the model? Are the samples large?

METHOD **Choosing the correct test from the wording**

1. Proportion / probability p: the variable counts "successes" out of n trials (defective items, voters in favour). Test $H_0 : p = p_0$ using $X \sim B(n, p_0)$ exactly (small n) or the normal approximation $\dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$ (large n).
2. Mean of a Poisson distribution: the variable counts events in a fixed interval of time or space (calls per hour, accidents per week). Test $H_0 : \lambda = \lambda_0$ using $X \sim \text{Po}(\lambda_0)$, with the normal approximation when λ_0 is large.
3. Population mean μ (one sample): the variable is a measurement (mass, length, time). Test $H_0 : \mu = \mu_0$ with $z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ (σ known, or n large using s for σ).
4. Difference of two means $\mu_1 - \mu_2$ (two independent samples): the question compares two groups, regions, machines or periods. Use

$$z = \frac{\bar{x}_1 - \bar{x}_2 - d_0}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}.$$



Decision tree for selecting the appropriate significance test from the wording of an exam question.

WORKED EXAMPLE 14 **Full exam-style walkthrough**

A factory manager suspects that Machine 1 produces rods whose mean length differs from that of Machine 2. Independent random samples give: Machine 1, $n_1 = 45$ rods, $\bar{x}_1 = 25.14$ cm, $s_1 = 0.42$ cm; Machine 2, $n_2 = 55$ rods, $\bar{x}_2 = 25.02$ cm, $s_2 = 0.38$ cm. Stating your hypotheses clearly, test the manager's suspicion at the 5% level of significance.

SOLUTION

Solution. Choice of test. Two independent samples, measured variable (length), both samples large: two-sample z-test for $\mu_1 - \mu_2$.
Hypotheses. "Differs" is two-sided: $H_0 : \mu_1 - \mu_2 = 0$; $H_1 : \mu_1 - \mu_2 \neq 0$; $\alpha = 5\%$.
Test statistic.

$$z = \frac{25.14 - 25.02}{\sqrt{\dfrac{0.42^2}{45} + \dfrac{0.38^2}{55}}} = \frac{0.12}{\sqrt{0.00392 + 0.002625}} = \frac{0.12}{\sqrt{0.006545}} = \frac{0.12}{0.0809} \approx 1.48.$$

Decision and conclusion. Since $|z| = 1.48 < 1.960$, we do not reject H_0 at the 5% level (p-value $\approx 2(0.0694) = 0.139 > 0.05$). There is insufficient evidence, at the 5% level, that the two machines produce rods of different mean lengths: the observed difference of 0.12 cm can reasonably be attributed to sampling fluctuation.

EXERCISE 9

A school inspector compares the mean scores of candidates in two divisions. Division 1: $n_1 = 70$ candidates, $\bar{x}_1 = 58.3$, $s_1 = 11.2$. Division 2: $n_2 = 65$ candidates, $\bar{x}_2 = 54.9$, $s_2 = 10.4$. Test at the 5% level whether the mean scores differ, and state whether your conclusion would change at the 1% level.

SOLUTION **Exercise 1**

$H_0 : \mu_1 - \mu_2 = 0$; $H_1 : \mu_1 - \mu_2 \neq 0$. Samples independent and large.

$$z = \frac{58.3 - 54.9}{\sqrt{\dfrac{11.2^2}{70} + \dfrac{10.4^2}{65}}} = \frac{3.4}{\sqrt{1.792 + 1.664}} = \frac{3.4}{1.859} \approx 1.83.$$

Since $1.83 < 1.960$, do not reject H_0 at the 5% level: there is no significant evidence of a difference between the mean scores. A fortiori, since $1.83 < 2.576$, the conclusion is unchanged at the 1% level.

EXERCISE 10

A nutritionist claims that a fortified flour increases the mean mass of 12-month-old children by more than 0.5 kg compared with the standard flour. Two independent groups of children are followed: fortified flour, $n_1 = 50$, mean gain $\bar{x}_1 = 3.1$ kg, $s_1 = 0.9$ kg; standard flour, $n_2 = 60$, mean gain $\bar{x}_2 = 2.4$ kg, $s_2 = 1.0$ kg. Test at the 5% level whether the additional gain exceeds 0.5 kg.

SOLUTION **Exercise 2**

The claim concerns $d_0 = 0.5$: $H_0 : \mu_1 - \mu_2 = 0.5$; $H_1 : \mu_1 - \mu_2 > 0.5$ (right tail), $\alpha = 5\%$. Samples independent and large.

$$z = \frac{3.1 - 2.4 - 0.5}{\sqrt{\dfrac{0.9^2}{50} + \dfrac{1.0^2}{60}}} = \frac{0.2}{\sqrt{0.0162 + 0.01667}} = \frac{0.2}{0.1814} \approx 1.10.$$

Since $z = 1.10 < 1.645$, do not reject H_0: at the 5% level there is insufficient evidence that the additional mean gain exceeds 0.5 kg.

KEY POINTS **Key points of the chapter**

- For independent samples, $E(\bar{X}_1 - \bar{X}_2) = \mu_1 - \mu_2$ and $\mathrm{Var}(\bar{X}_1 - \bar{X}_2) = \dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}$: variances add, standard deviations do not.
- For large n_1, n_2, the Central Limit Theorem gives $z = \dfrac{\bar{x}_1 - \bar{x}_2 - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \sim N(0,1)$ under H_0; replace σ_i^2 by s_i^2 when unknown.
- Critical values: two-tailed 1.960 (5%), 2.576 (1%); one-tailed 1.645 (5%), 2.326 (1%).
- Choose the test from the wording: proportion → binomial; Poisson mean → Poisson; one mean → one-sample z; comparing two means → two-sample z.
- Always conclude in the language of the problem, naming the level of significance, and never claim that H_0 has been proved.

8 INTEGRATION ACTIVITY

8.1 Integration Activity: Maize Yield Field Trial — Hypothesis Testing in Action

This activity consolidates Lessons 145–148 of the official syllabus. By the end, you should be able to:

- formulate a **null hypothesis** H_0 and an **alternative hypothesis** H_1 matched to a practical question, and choose between a **one-tailed** and a **two-tailed** test (Lesson 145);
- carry out a one-sample z-test for a **population mean** when σ is known, using the standardised statistic $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$ (Lesson 147);
- carry out a **two-sample z-test** for the difference between two means with large samples, justified by the Central Limit Theorem (Lesson 148);
- compute and interpret a **p-value**, and compare it with the significance level α;
- state a conclusion in plain language for a non-specialist audience, and explain the risk of a **Type I error**;
- discuss how the choice of α (here 5% versus 1%) affects the recommendation.

Work in groups of three or four. You will need statistical tables (or a calculator) giving $\Phi(z)$, the cumulative distribution function of $N(0,1)$.

8.2 Context

The Mbog Liaa farmers' cooperative, on the high plateau near Bafoussam (West Region), grows maize on family plots of roughly one hectare. Historical records using the traditional fertiliser show a stable mean yield $\mu_0 = 2.4$ tonnes per hectare, with a known plot-to-plot standard deviation $\sigma = 0.6$ tonnes per hectare.

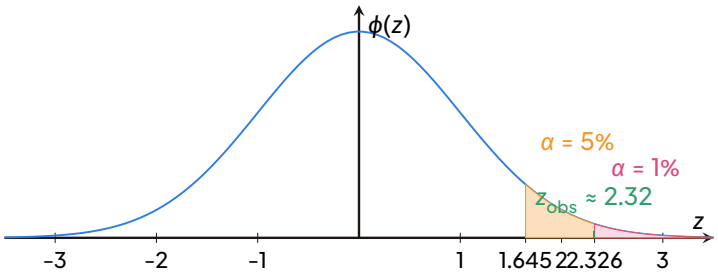
An agro-dealer in Bafoussam promotes a new, more expensive fertiliser, claiming it "significantly increases maize yield". Before committing members' money, the cooperative ran a field trial last rainy season:

- $n_1 = 40$ plots received the **new fertiliser**: sample mean yield $\bar{x}_1 = 2.62$ tonnes/ha;
- $n_2 = 50$ plots remained with the **old fertiliser** (control group): sample mean yield $\bar{x}_2 = 2.45$ tonnes/ha, sample standard deviation $s_2 = 0.65$ tonnes/ha.

The committee asks two questions:

1. Does the new fertiliser give a mean yield above the historical mean of 2.4 tonnes/ha?
2. Is the new fertiliser better than the old one, i.e. is the difference of means significantly positive?

The cooperative fears wasting money on a costly fertiliser that may not truly be better — your advisory note must mention the risk of concluding "it works" when it does not.



Standard normal density showing critical regions for one-tailed tests at 5% ($z > 1.645$) and 1% ($z > 2.326$), and the observed statistic $z_{obs} \approx 2.32$ for Test 1.

8.3 Tasks

1. **Setting up the first test.** Let μ be the true mean yield of plots using the new fertiliser. Write down H_0 and H_1 for a one-tailed test at the 5% level of the claim "the new fertiliser exceeds the historical mean". Justify the choice of a one-tailed test.

2. **Carrying out the first test.** Assuming $\sigma = 0.6$ is known, compute the test statistic $z = \dfrac{\bar{x}_1 - \mu_0}{\sigma/\sqrt{n_1}}$. State the critical value at the 5% level (one-tailed) and conclude.

3. ***p*-value of the first test.** Compute the p-value $p = P(Z \geq z_{obs})$ where $Z \sim N(0,1)$. Interpret this number in one sentence for the committee.

4. **Setting up the second test.** Let μ_1 and μ_2 be the true mean yields under the new and old fertilisers. Write H_0 and H_1 for testing, at the 5% level, whether the new fertiliser outperforms the old one.

5. **Carrying out the second test.** Explain why, with $n_1 = 40$ and $n_2 = 50$, the Central Limit Theorem justifies a z-test, and why s_2 may be used in place of the unknown σ_2. Using $\sigma_1 = 0.6$ (known) and $s_2 = 0.65$, compute
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{s_2^2}{n_2}}},$$
state the critical region, and conclude.

6. ***p*-value of the second test.** Compute the p-value and compare it with $\alpha = 0.05$.

7. **Advisory note.** In at most ten lines, write a note to the cooperative's president summarising both conclusions in plain language, and explaining in one or two sentences what a Type I error would mean here and its probability under H_0.

8. **Sensitivity to α.** Repeat the decision step of both tests at the 1% level (critical value 2.326). Does the recommendation change? Discuss in your group: what are the consequences, for the farmers, of choosing a smaller α?

8.4 Model Answers and Explanations

1. Hypotheses for the first test (Lesson 145, 147). We test the historical mean against an increase:

$$H_0 : \mu = 2.4 \qquad \text{against} \qquad H_1 : \mu > 2.4 \quad (\text{one-tailed}, \alpha = 0.05).$$

A one-tailed test is appropriate because the cooperative only wants to switch if the new fertiliser is better; a decrease would be of no practical interest. The alternative hypothesis reflects the dealer's claim of an increase.

2. First test statistic (Lesson 147). Under H_0, the sample mean $\bar{X}_1$ follows a normal distribution (exactly, since the population is normal or approximately by CLT with $n_1 = 40$):

$$\bar{X}_1 \sim N\left(2.4,\ \frac{0.6^2}{40}\right).$$

The standardised test statistic is

$$z = \frac{2.62 - 2.4}{0.6/\sqrt{40}} = \frac{0.22}{0.094868\ldots} \approx 2.319.$$

The critical value for a one-tailed test at 5% is $z_{0.05} = 1.645$. Since $2.319 > 1.645$, the observed value lies in the critical region. We **reject** H_0 at the 5% level: there is significant evidence that the new fertiliser exceeds the historical mean.

3. p-value of the first test.

$$p = P(Z \geq 2.319) = 1 - \Phi(2.319) \approx 1 - 0.9898 = 0.0102.$$

Since $p \approx 0.0102 < 0.05$, H_0 is rejected. Interpretation: If the new fertiliser were truly no better than the historical mean (2.4 t/ha), a sample mean as high as 2.62 t/ha (or higher) would occur in only about 1% of similar trials of 40 plots.

4. Hypotheses for the second test (Lesson 145, 148).

$$H_0 : \mu_1 - \mu_2 = 0 \qquad \text{against} \qquad H_1 : \mu_1 - \mu_2 > 0 \quad (\alpha = 0.05).$$

This is a one-tailed test because the cooperative only cares whether the new fertiliser is superior to the old one.

5. Second test statistic (Lesson 148). Both sample sizes are large ($n_1 = 40$, $n_2 = 50$, both > 30). By the Central Limit Theorem, the sampling distribution of $\bar{X}_1 - \bar{X}_2$ is approximately normal regardless of the underlying yield distributions, provided the plots are independent and randomly assigned. The population standard deviation for the new fertiliser is known ($\sigma_1 = 0.6$); for the old fertiliser it is unknown, but with $n_2 = 50$ the sample standard deviation $s_2 = 0.65$ is a reliable estimate of σ_2. Under H_0:

$$z = \frac{2.62 - 2.45}{\sqrt{\dfrac{0.6^2}{40} + \dfrac{0.65^2}{50}}} = \frac{0.17}{\sqrt{0.009 + 0.00845}} = \frac{0.17}{\sqrt{0.01745}} \approx \frac{0.17}{0.1321} \approx 1.287.$$

The critical value at 5% (one-tailed) is 1.645. Since $1.287 < 1.645$, the observed value does not lie in the critical region. We **do not reject** H_0 at the 5% level: the trial does not provide significant evidence that the new fertiliser beats the old one.

6. p-value of the second test.

$$p = P(Z \geq 1.287) = 1 - \Phi(1.287) \approx 1 - 0.9015 = 0.0985.$$

Since $p \approx 0.0985 > 0.05$, this confirms the non-rejection of H_0.

WARNING **Apparent disagreement between the two tests**

The first test compares the new fertiliser with the historical mean (2.4 t/ha); the second compares it with this season's control group (2.45 t/ha). The control group's mean happens to be slightly above the historical figure, possibly due to favourable weather or soil conditions this season. The two questions are different, so different conclusions are not contradictory. This illustrates why the choice of the null hypothesis must match the practical question.

7. Advisory note (model).

KEY POINTS **Note to the President of Mbog Liaa Cooperative**

Our statistical analysis of the field trial yields the following conclusions. (i) Compared with the cooperative's historical average of 2.4 tonnes/ha, the 40 plots using the new fertiliser performed significantly better ($p \approx 0.01$). (ii) However, compared directly with the 50 control plots using the old fertiliser this same season, the advantage of the new fertiliser is not statistically significant ($p \approx 0.10$): the observed gap of 0.17 tonne/ha could plausibly be due to random variation between plots. There is a small risk (at most 5%, the significance level) of a "false alarm" — declaring the new fertiliser better when it is not (Type I error). Recommendation: before adopting the more expensive fertiliser for all members, repeat the trial next season on a larger number of plots to confirm the advantage.

8. Sensitivity to α (Lesson 145). At the 1% level the critical value is $z_{0.01} = 2.326$.

Test	Statistic	Decision at 5%	Decision at 1%
New vs. historical mean	$z \approx 2.32$	Reject H_0	Do not reject (2.32 < 2.326)
New vs. old (two-sample)	$z \approx 1.29$	Do not reject	Do not reject

At the 1% level, neither test rejects H_0, so the cautious recommendation would be "no proven improvement". Lowering α reduces the risk of a Type I error (wasting members' money on an ineffective product) but increases the risk of a **Type II error** (missing a genuinely better fertiliser). Choosing α is therefore a real economic decision for the cooperative, not merely a mathematical one — and increasing the sample size (larger n) is the only way to reduce both risks simultaneously.

DID YOU KNOW?

Type I and Type II errors in agricultural trials

In Cameroonian agricultural extension, a Type I error means recommending an input (fertiliser, seed variety) that does not actually improve yields — farmers spend scarce cash for no gain. A Type II error means failing to recommend a truly better input — farmers miss out on higher income. The GCE syllabus expects you to articulate this trade-off in context. The standard 5% level is a convention; in high-stakes decisions (e.g. new vaccine, structural safety) $\alpha = 1\%$ or even 0.1% may be used.

9 METHODS AND WORKED EXAMPLES

10 HYPOTHESIS TESTING

10.1 Setting up hypotheses and identifying the test

METHOD **Choosing H_0, H_1 and the type of test**

1. Read the question and identify the **claim** to be tested. The **null hypothesis** H_0 always contains an equality: $H_0 : \theta = \theta_0$ (the status quo, "no change", "no difference").
2. The **alternative hypothesis** H_1 expresses what the investigator suspects:
   - $H_1 : \theta \neq \theta_0$ (**two-tailed** test: "has changed", "differs from");
   - $H_1 : \theta > \theta_0$ (one-tailed, right: "has increased", "is greater");
   - $H_1 : \theta < \theta_0$ (one-tailed, left: "has decreased", "is less").
3. Choose the **test statistic** X and state its distribution under H_0: binomial $B(n, p_0)$ for a probability, Poisson $\mathrm{Po}(\lambda_0)$ for a Poisson mean, $N\left(\mu_0, \sigma^2/n\right)$ for a sample mean.
4. Fix the **significance level** α (e.g. 5%). For a two-tailed test, split it: $\alpha/2$ in each tail.
5. Conclude in context: "reject H_0" (evidence against the claim) or "do not reject H_0" (insufficient evidence) — never "prove".

WORKED EXAMPLE 15 **Identifying hypotheses and tails**

A Yaoundé bottling company claims that the mean volume of its "Top" soda bottles is 35 cL. (a) The quality controller suspects the machine under-fills the bottles. (b) A consumer association suspects the mean volume differs from 35 cL. In each case, state H_0 and H_1 and the type of test.

SOLUTION

Let μ be the true mean volume (in cL) of bottles from the machine.
(a) The controller suspects under-filling, i.e. $\mu < 35$. Hence

$$H_0 : \mu = 35 \qquad \text{against} \qquad H_1 : \mu < 35.$$

This is a one-tailed (left-tail) test: only small values of the sample mean count as evidence against H_0.

(b) The association suspects the volume differs from 35 cL, in either direction:

$$H_0 : \mu = 35 \qquad \text{against} \qquad H_1 : \mu \neq 35.$$

This is a two-tailed test: both unusually small and unusually large sample means count as evidence, and the significance level must be shared between the two tails.

10.2 Testing a probability (binomial model)

METHOD **Significance test for a probability p**

1. State $H_0 : p = p_0$ and H_1 (one- or two-tailed). Under H_0, the count of successes is $X \sim B(n, p_0)$.
2. The **p-value** reflex: compute the probability, under H_0, of a result at least as extreme as the observed value x:
   - left tail: $P(X \leq x)$; right tail: $P(X \geq x)$;
   - two-tailed: double the appropriate tail probability (or compare each tail with $\alpha/2$).
3. Decision rule: reject H_0 if the p-value $< \alpha$ (for two-tailed, compare with $\alpha/2$ per tail).
4. Alternatively, find the **critical region**: the set of values of X with total probability $\leq \alpha$ under H_0; reject H_0 if x lies in it.
5. Conclude in the language of the question, quoting the level of significance.

WORKED EXAMPLE 16 **Testing a probability — one tail**

A seed merchant in Bafoussam claims that 80% of his cabbage seeds germinate. A farmer plants 20 seeds and only 12 germinate. Test, at the 5% level of significance, whether the merchant's claim is too optimistic.

SOLUTION

Step 1 — Hypotheses. Let p be the germination probability. The farmer suspects p is smaller than claimed:

$$H_0 : p = 0.8 \qquad H_1 : p < 0.8 \quad \text{(one-tailed, left).}$$

Step 2 — Model under H_0. If X is the number of seeds that germinate out of 20, then under

H_0,

$$X \sim B(20,\ 0.8).$$

Observed value: $x = 12$.
Step 3 — p-value. We compute the probability of 12 or fewer germinations under H_0. It is easier to count failures: let $Y = 20 - X \sim B(20, 0.2)$; then $X \le 12 \iff Y \ge 8$.

$$\begin{aligned} P(Y \ge 8) &= 1 - P(Y \le 7) \\ &= 1 - \sum_{k=0}^{7} \binom{20}{k} (0.2)^k (0.8)^{20-k} \\ &= 1 - 0.9679 = 0.0321. \end{aligned}$$

Step 4 — Decision. Since $0.0321 < 0.05$, the observed result is in the lower 5% tail: we reject H_0.
Step 5 — Conclusion. At the 5% level of significance there is sufficient evidence that the germination rate is less than 80%: the merchant's claim appears too optimistic.

WORKED EXAMPLE 17 **Finding a critical region**

A coin is suspected of being biased towards heads. It is tossed 15 times. Find the critical region for a test of $H_0 : p = \frac{1}{2}$ against $H_1 : p > \frac{1}{2}$ at the 5% level, and state the actual significance level of the test.

SOLUTION

Under H_0, $X \sim B(15, 0.5)$, where X counts heads. We seek the smallest c with $P(X \ge c) \le 0.05$. Using symmetry of $B(15, 0.5)$ and cumulative tables:

$$\begin{aligned} P(X \ge 11) = P(X \le 4) &= \sum_{k=0}^{4} \binom{15}{k} (0.5)^{15} \\ &= \frac{1 + 15 + 105 + 455 + 1365}{32768} = \frac{1941}{32768} \approx 0.0592, \\ P(X \ge 12) = P(X \le 3) &= \frac{1 + 15 + 105 + 455}{32768} = \frac{576}{32768} \approx 0.0176. \end{aligned}$$

Since $0.0592 > 0.05$ but $0.0176 \le 0.05$, the critical region is

$$X \ge 12, \qquad \text{i.e. } X \in \{12, 13, 14, 15\}.$$

The actual significance level is $P(X \ge 12 \mid p = 0.5) \approx 0.0176$, i.e. about 1.76% — necessarily below 5% because the binomial distribution is discrete.

10.3 Testing the mean of a Poisson distribution

METHOD **Significance test for a Poisson mean λ**

1. State $H_0 : \lambda = \lambda_0$ and H_1. Under H_0 the count over the observed period is $X \sim \text{Po}(\lambda_0)$.
2. **Reflex — adjust the rate**: if the observation covers a different period/area from the one defining λ_0, scale λ proportionally (e.g. rate per hour, observe for 3 hours: $\lambda = 3\lambda_0$).
3. Compute the tail probability under H_0: left tail $P(X \le x)$, right tail $P(X \ge x) = 1 - P(X \le x-1)$, using $P(X = k) = e^{-\lambda_0}\lambda_0^k / k!$.
4. Compare with α (or $\alpha/2$ for two tails), decide, and conclude in context.

WORKED EXAMPLE 18 **Testing a Poisson mean**

Calls arrive at the call centre of a Douala telephone company at an average rate of 4 per hour. After a publicity campaign, the centre receives 11 calls in one hour. Test at the 5% level whether the mean rate has increased.

SOLUTION

Step 1 — Hypotheses. Let λ be the mean number of calls per hour.

$$H_0 : \lambda = 4 \qquad H_1 : \lambda > 4 \quad \text{(one-tailed, right)}.$$

Step 2 — Model. Under H_0, the number of calls in one hour is $X \sim \text{Po}(4)$. Observed: $x = 11$.
Step 3 — p-value. We need $P(X \ge 11) = 1 - P(X \le 10)$ under Po(4). Summing $e^{-4}4^k/k!$:

$$P(X \le 10) = e^{-4}\sum_{k=0}^{10}\frac{4^k}{k!} \approx 0.9972,$$

so

$$P(X \ge 11) \approx 1 - 0.9972 = 0.0028.$$

Step 4 — Decision. $0.0028 < 0.05$: reject H_0.
Step 5 — Conclusion. At the 5% level there is strong evidence that the mean rate of calls has increased above 4 per hour; the campaign appears to have had an effect.

Remark. If instead we had observed 9 calls, then $P(X \ge 9) = 1 - P(X \le 8) \approx 1 - 0.9786 = 0.0214 < 0.05$, still significant; but 8 calls gives $P(X \ge 8) \approx 0.0511 > 0.05$, not significant. The critical region at 5% is $X \ge 9$.

10.4 Testing a population mean

METHOD **Test for a mean μ — σ known or large sample**

1. State $H_0 : \mu = \mu_0$ and H_1. Identify σ (known) or estimate it by s if n is large ($n \geq 30$), invoking the **Central Limit Theorem**.
2. The test statistic is the standardised sample mean:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \quad \sim \quad N(0,1) \text{ under } H_0.$$

3. Find the **critical value(s)**: ±1.645 (one tail, 5%), ±1.96 (two tails, 5%), ±2.326 (one tail, 1%), ±2.576 (two tails, 1%).
4. Reject H_0 if z falls beyond the critical value in the direction of H_1; otherwise do not reject.
5. Conclude in context, mentioning the significance level.

WORKED EXAMPLE 19 **Testing a mean, σ known — two tails**

A machine fills packets of sugar; the masses are normally distributed with standard deviation 5 g. The machine is set to deliver a mean of 500 g. A random sample of 25 packets has mean mass 502.8 g. Test at the 5% level whether the mean mass differs from 500 g.

SOLUTION

Step 1 — Hypotheses.

$$H_0 : \mu = 500 \qquad H_1 : \mu \neq 500 \quad \text{(two-tailed)}.$$

Step 2 — Statistic. Under H_0, $\bar{X} \sim N\left(500, \frac{5^2}{25}\right) = N(500, 1)$, so $\sigma/\sqrt{n} = 1$ g.

$$z = \frac{502.8 - 500}{5/\sqrt{25}} = \frac{2.8}{1} = 2.8.$$

Step 3 — Critical values. Two-tailed at 5%: ±1.96.
Step 4 — Decision. Since $|z| = 2.8 > 1.96$, we reject H_0.
Step 5 — Conclusion. At the 5% level there is significant evidence that the mean mass is no longer 500 g; the machine needs readjustment (indeed the evidence points to overfilling).

WORKED EXAMPLE 20 **Testing a mean, σ unknown, large sample**

The mean score of candidates in a national mock examination has historically been 52 marks. A random sample of 60 candidates this year gives a mean of 54.3 marks with standard deviation 11.5 marks. Test at the 1% level whether the mean has increased.

SOLUTION

Step 1 — Hypotheses.

$$H_0 : \mu = 52 \qquad H_1 : \mu > 52 \quad \text{(one-tailed, right).}$$

Step 2 — Statistic. Here σ is unknown, but $n = 60$ is large, so by the Central Limit Theorem we may use $s = 11.5$ in place of σ:

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{54.3 - 52}{11.5/\sqrt{60}} = \frac{2.3}{1.4846} \approx 1.549.$$

Step 3 — Critical value. One-tailed at 1%: $z_{\text{crit}} = 2.326$.
Step 4 — Decision. Since $1.549 < 2.326$, we do not reject H_0.
Step 5 — Conclusion. At the 1% level there is insufficient evidence that the mean mark has increased; the observed improvement can reasonably be attributed to sampling fluctuation.

10.5 Testing the difference between two means (large samples)

METHOD **Two-sample z-test for $\mu_1 - \mu_2$**

1. State $H_0 : \mu_1 - \mu_2 = 0$ (i.e. $\mu_1 = \mu_2$) and H_1 (usually $\mu_1 \neq \mu_2$ or $\mu_1 > \mu_2$).
2. For large independent samples ($n_1, n_2 \geq 30$), the CLT gives, under H_0:

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}} \quad \sim \quad N(0,1).$$

3. **Reflex**: the standard error of a difference adds the two variances $\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}$ — never subtract them.
4. Compare $|z|$ (or z, one tail) with the critical value, decide, and conclude in context.

WORKED EXAMPLE 21 **Difference of two means — two tails**

An inspector compares the yields of two maize varieties. Variety A: 50 farms, mean yield 2.45 tonnes/ha, standard deviation 0.60. Variety B: 60 farms, mean yield 2.28 tonnes/ha, standard deviation 0.55. Test at the 5% level whether the two varieties differ in mean yield.

SOLUTION

Step 1 — Hypotheses. Let μ_A, μ_B be the true mean yields.

$$H_0 : \mu_A = \mu_B \qquad H_1 : \mu_A \neq \mu_B \quad \text{(two-tailed).}$$

Step 2 — Standard error.

$$\text{SE} = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = \sqrt{\frac{0.60^2}{50} + \frac{0.55^2}{60}} = \sqrt{0.0072 + 0.00504} = \sqrt{0.01224} \approx 0.1106.$$

Step 3 — Test statistic.

$$z = \frac{\bar{x}_A - \bar{x}_B}{\text{SE}} = \frac{2.45 - 2.28}{0.1106} = \frac{0.17}{0.1106} \approx 1.537.$$

Step 4 — Decision. Two-tailed at 5%: critical values ±1.96. Since $|z| = 1.537 < 1.96$, we do not reject H_0.
Step 5 — Conclusion. At the 5% level the data do not provide sufficient evidence of a difference in mean yield between the two varieties; the observed gap of 0.17 tonnes/ha is compatible with ordinary sampling variability.

WORKED EXAMPLE 22 **Difference of two means — one tail, with p-value**

A new teaching method is tried in the West Region. Group 1 (new method): 80 students, mean 61.2, standard deviation 9.8. Group 2 (traditional method): 100 students, mean 58.5, standard deviation 10.4. Test at the 1% level whether the new method produces a higher mean score, and give the p-value.

SOLUTION

Step 1 — Hypotheses.

$$H_0 : \mu_1 = \mu_2 \qquad H_1 : \mu_1 > \mu_2 \quad \text{(one-tailed, right).}$$

Step 2 — Standard error.

$$\text{SE} = \sqrt{\frac{9.8^2}{80} + \frac{10.4^2}{100}} = \sqrt{\frac{96.04}{80} + \frac{108.16}{100}} = \sqrt{1.2005 + 1.0816} = \sqrt{2.2821} \approx 1.5107.$$

Step 3 — Test statistic.

$$z = \frac{61.2 - 58.5}{1.5107} = \frac{2.7}{1.5107} \approx 1.787.$$

Step 4 — Decision. One-tailed at 1%: $z_{\text{crit}} = 2.326$. Since $1.787 < 2.326$, we do not reject H_0.

Step 5 — p-value. $p = P(Z \geq 1.787) = 1 - \Phi(1.787) \approx 1 - 0.9630 = 0.0370$, i.e. about 3.7%. The result would have been significant at the 5% level, but not at the stricter 1% level.

Conclusion. At the 1% level there is insufficient evidence that the new method raises the mean score, although the data are suggestive ($p \approx 3.7\%$): a larger study would be worthwhile.

10.6 Errors and interpretation

METHOD **Type I and Type II errors**

1. A **Type I error**: rejecting H_0 when it is true. Its probability is the significance level α (for discrete tests, the actual size of the critical region).
2. A **Type II error**: failing to reject H_0 when H_1 is true. Its probability β is computed at a specified alternative value: find the acceptance region under H_0, then evaluate the probability of that region under the alternative distribution.
3. The **power** of the test is $1 - \beta$.
4. Increasing n (or widening the critical region) reduces β; reducing α increases β. There is always a trade-off.

WORKED EXAMPLE 23 **Computing error probabilities**

A test of $H_0 : \mu = 50$ against $H_1 : \mu < 50$ uses a sample of $n = 36$ from $N(\mu, 12^2)$, rejecting H_0 when $\bar{X} < 46.71$. (a) Find the significance level. (b) Find the probability of a Type II error and the power when the true mean is $\mu = 45$.

SOLUTION

Under H_0, $\bar{X} \sim N\left(50, \frac{12^2}{36}\right) = N(50, 4)$, so the standard error is 2.

(a) Significance level (Type I error).

$$\alpha = P(\bar{X} < 46.71 \mid \mu = 50) = P\left(Z < \frac{46.71 - 50}{2}\right) = P(Z < -1.645) = 0.05.$$

The test is at the 5% level (the critical value 46.71 was chosen exactly for this).

(b) Type II error at $\mu = 45$. Now $\bar{X} \sim N(45, 4)$. A Type II error occurs when we accept H_0, i.e.

$\bar{X} \geq 46.71$:

$$\beta = P(\bar{X} \geq 46.71 \mid \mu = 45) = P\left(Z \geq \frac{46.71 - 45}{2}\right) = P(Z \geq 0.855).$$

From tables, $\Phi(0.855) \approx 0.8037$, so $\beta \approx 0.196$.
The power of the test at $\mu = 45$ is

$$1 - \beta \approx 0.804.$$

Interpretation: if the true mean is really 45, this test detects it about 80% of the time; in about 20% of samples we would wrongly fail to reject H_0.

WARNING **Frequent traps at the GCE**

- Writing H_1 with an equality sign, or putting the sample value ($\bar{x}$) into the hypotheses instead of the population parameter μ or p.
- For a two-tailed test, comparing the one-sided p-value with α instead of $\alpha/2$.
- Forgetting to scale the Poisson rate to the observed time interval.
- Subtracting instead of adding the variances in the standard error of $\bar{X}_1 - \bar{X}_2$.
- Saying "H_0 is true" instead of "there is insufficient evidence to reject H_0".
- Forgetting that for discrete distributions (binomial, Poisson) the actual significance level is the exact probability of the critical region, usually smaller than the nominal α.

11 EXERCISES

11.1 I check my knowledge

EXERCISE 11 **MCQ: the language of testing**

For each question, choose the single correct answer and justify briefly.

1. In a significance test, the null hypothesis H_0 is:
   (a) the hypothesis we suspect to be true and wish to prove;
   (b) the hypothesis of "no change / no effect", rejected only if the data provide strong evidence against it;
   (c) the hypothesis that is always accepted at the end of the test;
   (d) the hypothesis that the sample mean equals the population mean.
2. The significance level α of a test is:
   (a) the probability of accepting H_0 when H_0 is true;
   (b) the probability of rejecting H_0 when H_0 is true;
   (c) the probability of rejecting H_0 when H_1 is true;
   (d) always equal to 5%.
3. A Type II error consists in:
   (a) rejecting H_0 when H_0 is true;
   (b) accepting H_0 when H_0 is false;
   (c) choosing a one-tailed test instead of a two-tailed test;
   (d) using the wrong significance level.
4. For a two-tailed z-test at the 5% level, the critical values are:
   (a) ± 1.645;
   (b) ± 1.960;
   (c) ± 2.326;
   (d) ± 2.576.

EXERCISE 12 **True or false?**

State whether each assertion is true or false, and justify your answer in one or two sentences.

1. If the test statistic falls in the critical region, we have proved that H_0 is false.
2. Decreasing the significance level from 5% to 1% makes it harder to reject H_0.
3. For a large sample, the Central Limit Theorem allows us to use a normal approximation for the sample mean even when the population is not normal.
4. The p-value is the probability that H_0 is true given the observed data.
5. A 95% confidence interval that does not contain the value claimed by H_0 leads to rejection of H_0 in a two-tailed test at the 5% level.

EXERCISE 13 **Key definitions**

1. Define, in your own words, the terms: **null hypothesis**, **alternative hypothesis**, **test statistic**, **critical region** and **significance level**.
2. Explain the difference between a **one-tailed** test and a **two-tailed** test, and give one concrete situation (from everyday life in Cameroon) appropriate to each.
3. Distinguish carefully between a Type I error and a Type II error, illustrating each with the situation: “testing whether a new malaria treatment is more effective than the standard one”.

EXERCISE 14 **Direct calculations**

1. Let $X \sim B(10, p)$. We test $H_0 : p = 0.5$ against $H_1 : p > 0.5$. Compute $P(X \geq 8 \mid p = 0.5)$ and $P(X \geq 9 \mid p = 0.5)$. Which of the regions $\{X \geq 8\}$ and $\{X \geq 9\}$ gives a significance level closest to 5%?
2. Let $Y \sim P(\lambda)$ with $\lambda = 4$. Compute $P(Y \geq 8)$ (you may use $P(Y \geq 8) = 1 - P(Y \leq 7)$ and the cumulative Poisson table). Would observing $Y = 8$ lead to rejection of $H_0 : \lambda = 4$ against $H_1 : \lambda > 4$ at the 5% level?
3. A sample of $n = 64$ observations gives $\bar{x} = 52$ and $s = 8$. Compute the value of the test statistic $z = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ for testing $H_0 : \mu = 50$.

11.2 I practise

EXERCISE 15 **The lottery claim**

A lottery company claims that 1 ticket in 10 wins a prize. A students' association of a school in Bamenda buys 50 tickets and only 2 of them win a prize. Let X be the number of winning tickets, and assume $X \sim B(50, p)$.

1. State the null hypothesis H_0 and the alternative hypothesis H_1 for testing whether the claimed probability is overstated.
2. Using a suitable normal approximation (checking its validity), test the claim at the 5% level of significance.
3. State your conclusion clearly in the context of the problem.

EXERCISE 16 **The call centre**

Calls arrive at a call centre in Douala at a claimed mean rate of 6 per hour, and the number of calls per hour is modelled by a Poisson distribution. In one observed hour, 12 calls arrive.

1. Write down H_0 and H_1 for testing whether the mean rate has increased.
2. Using the cumulative Poisson tables, compute $P(X \geq 12)$ when $\lambda = 6$.
3. Carry out the test at the 1% level of significance and interpret the result in context.

EXERCISE 17 **Sachet water**

A sachet-water factory in Yaoundé claims that the mean content of its sachets is 500 ml. A random sample of 36 sachets gives a sample mean $\bar{x} = 496$ ml and a sample standard deviation $s = 9$ ml.

1. State H_0 and H_1 for testing whether the true mean is below 500 ml.
2. Compute the test statistic and carry out the test at the 5% level.
3. Compute the p-value of the observed sample mean and compare it with $\alpha = 0.05$.
4. What practical advice would you give to the factory's quality-control manager?

EXERCISE 18 **Bags of cement**

The mean mass of bags of cement produced by a factory is claimed to be 50 kg, with a

known standard deviation $\sigma = 1.2$ kg. A random sample of 25 bags gives a sample mean $\bar{x} = 50.7$ kg.

1. Test, at the 5% level of significance, the two-tailed hypotheses $H_0 : \mu = 50$ against $H_1 : \mu \neq 50$.
2. Construct a 95% confidence interval for μ and verify that it leads to the same conclusion as the test.
3. Explain in one sentence why the confidence interval and the two-tailed test must agree.

11.3 I prepare for the GCE

EXERCISE 19 **Two schools at the mock GCE** **(12 marks)**

In a mock GCE Advanced Level examination, 60 candidates from school A obtain a mean mark of 58.4 with sample standard deviation 11.2, while 80 candidates from school B obtain a mean mark of 55.1 with sample standard deviation 12.5. The marks in each school may be assumed independent.

1. State suitable null and alternative hypotheses for testing whether the mean mark of school A is significantly higher than that of school B. (2)
2. Justify the use of a normal distribution for the difference of the two sample means, quoting the relevant theorem. (2)
3. Compute the standard error of $\bar{x}_A - \bar{x}_B$ and the value of the test statistic. (4)
4. Carry out the test at the 5% level of significance and state your conclusion in context. (3)
5. State one assumption underlying your analysis. (1)

EXERCISE 20 **Finding a critical region** **(10 marks)**

A seed merchant claims that the proportion p of germinating seeds in a batch is 0.4. A buyer suspects that the germination rate is in fact lower. Let $X \sim B(20, p)$ be the number of germinating seeds in a random sample of 20 seeds. We test $H_0 : p = 0.4$ against $H_1 : p < 0.4$.

1. Explain why the critical region has the form $X \le c$ for some integer c. (2)
2. Using the binomial cumulative distribution, compute $P(X \le 4 \mid p = 0.4)$ and $P(X \le 3 \mid p = 0.4)$. (4)

3. Hence determine the critical region whose significance level is as close as possible to 5%, and state the actual significance level of the test. (3)

4. The buyer observes 4 germinating seeds. What is his conclusion? (1)

EXERCISE 21 **Malaria cases after a flood** **(14 marks)**

A clinic in the Far North Region records malaria cases daily; the number of cases per day is modelled by a Poisson distribution with mean 3. After a flood, 9 cases are recorded on a single day. The medical officer wishes to know whether the mean daily number of cases has increased.

1. State, with justification, the null and alternative hypotheses. (3)

2. Using cumulative Poisson probabilities, compute the p-value $P(X \geq 9 \mid \lambda = 3)$. (4)

3. Carry out the test at the 1% level of significance. (3)

4. Write a clear conclusion addressed to the medical officer, in the context of the problem. (2)

5. Identify which type of error (Type I or Type II) could have been committed in your conclusion, and describe in one sentence what that error would mean for the clinic. (2)

12 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ: the language of testing**

1. Answer (b). H_0 is the hypothesis of "no change / no effect"; it is given the benefit of the doubt and is rejected only when the data provide strong evidence against it. Answer (a) is wrong because we never prove H_0; (c) is wrong because H_0 may be rejected; (d) confuses a sample result with a population claim.
2. Answer (b). By definition $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}) = P(\text{Type I error})$. Answer (a) is $1 - \alpha$; (c) is the power of the test; (d) is false since α is chosen by the experimenter (1%, 5%, 10%, …).
3. Answer (b). A Type II error is accepting H_0 when H_0 is in fact false. Answer (a) describes a Type I error; (c) and (d) are methodological choices, not errors of conclusion.
4. Answer (b). For a two-tailed test at the 5% level we need $P(|Z| > z) = 0.05$, i.e. $P(Z > z) = 0.025$ in each tail, giving $z = 1.960$. The value 1.645 is for a one-tailed 5% test; 2.326 and 2.576 correspond to 1% one-tailed and two-tailed tests respectively.

SOLUTION **Exercise 2 — True or false?**

1. False. Rejecting H_0 means the data are unlikely under H_0 (probability at most α), not impossible. There is always a probability α of a Type I error, so nothing is "proved" in the mathematical sense.
2. True. Lowering α shrinks the critical region (e.g. critical value 1.645 becomes 2.326 for a one-tailed test), so stronger evidence is required before H_0 is rejected.
3. True. The Central Limit Theorem states that for large n, $\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$ whatever the shape of the parent population, which justifies the normal-based test.
4. False. The p-value is $P(\text{result at least as extreme as observed} \mid H_0 \text{ true})$: it is computed assuming H_0 true, and says nothing about the probability of H_0 itself.
5. True. A 95% confidence interval contains exactly those values μ_0 that would not be rejected by a two-tailed test at the 5% level; hence a value outside the interval is rejected. The two procedures are mathematically equivalent.

SOLUTION **Exercise 3 — Key definitions**

1. - **Null hypothesis** H_0: the default statement of "no change / no effect" about a population parameter, assumed true unless the data strongly contradict it.
   - **Alternative hypothesis** H_1: the statement we suspect to be true; it is accepted only if the evidence against H_0 is strong enough.
   - **Test statistic**: a quantity computed from the sample (e.g. $z = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$) whose distribution under H_0 is known and which measures how far the data are from H_0.
   - **Critical region**: the set of values of the test statistic which are so unlikely under H_0 that their occurrence leads to rejection of H_0.
   - **Significance level** α: the probability, fixed in advance, of rejecting H_0 when it is actually true (probability of a Type I error).
2. In a **one-tailed** test H_1 specifies a direction ($\mu > \mu_0$ or $\mu < \mu_0$) and the critical region lies in a single tail; in a **two-tailed** test $H_1 : \mu \neq \mu_0$ and the critical region is split between both tails. One-tailed example: a trader in Douala suspects a supplier's bags of rice weigh less than the claimed 50 kg. Two-tailed example: a machine filling sachet water must deliver exactly 500 ml; any deviation, above or below, requires adjustment.
3. - Type I error: rejecting H_0 when it is true. Here: concluding the new malaria treatment is more effective when in reality it is not — a useless (possibly expensive) drug is adopted.
   - Type II error: accepting H_0 when it is false. Here: concluding the new treatment is no better when in reality it is more effective — patients are denied a better treatment.

SOLUTION **Exercise 4 — Direct calculations**

1. Under H_0, $X \sim B(10, 0.5)$, so $P(X = k) = \binom{10}{k}\dfrac{1}{2^{10}} = \dfrac{\binom{10}{k}}{1024}$.

$$P(X \geq 8) = \frac{\binom{10}{8} + \binom{10}{9} + \binom{10}{10}}{1024} = \frac{45 + 10 + 1}{1024} = \frac{56}{1024} \approx 0.0547.$$

$$P(X \geq 9) = \frac{\binom{10}{9} + \binom{10}{10}}{1024} = \frac{11}{1024} \approx 0.0107.$$

The region $\{X \geq 8\}$ has level 5.47%, the region $\{X \geq 9\}$ has level 1.07%. Hence $\{X \geq 8\}$ gives the significance level closest to 5%.

2. From cumulative Poisson tables with $\lambda = 4$: $P(Y \le 7) = 0.9489$, so

$$P(Y \ge 8) = 1 - 0.9489 = 0.0511.$$

Since $0.0511 > 0.05$, observing $Y = 8$ does not fall in the 5% critical region: we do not reject $H_0 : \lambda = 4$ at the 5% level. (Note how close the decision is!)

3. With $n = 64$, $\bar{x} = 52$, $s = 8$, $\mu_0 = 50$:

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{52 - 50}{8/\sqrt{64}} = \frac{2}{8/8} = \frac{2}{1} = 2.$$

The test statistic takes the value $z = 2.00$.

SOLUTION **Exercise 5 — The lottery claim**

1. The company claims $p = 0.1$; the association suspects the probability is smaller. Hence

$$H_0 : p = 0.1 \quad \text{against} \quad H_1 : p < 0.1 \quad \text{(one-tailed test)}.$$

2. Under H_0, $X \sim B(50, 0.1)$ with $np = 5$ and $np(1-p) = 4.5$. Since $np = 5 \ge 5$ and $n(1-p) = 45 \ge 5$, the normal approximation is acceptable (borderline for $np = 5$, but standard in GCE context):

$$X \approx N(5,\ 4.5), \qquad \sigma = \sqrt{4.5} \approx 2.121.$$

Applying the continuity correction, the observed value $x = 2$ is replaced by 2.5:

$$z = \frac{2.5 - 5}{2.121} = \frac{-2.5}{2.121} \approx -1.18.$$

(Without continuity correction, $z = \dfrac{2 - 5}{2.121} \approx -1.41$; the conclusion is the same.) The critical value for a one-tailed test at the 5% level is -1.645. Since $-1.18 > -1.645$, the test statistic does not lie in the critical region.

3. Conclusion: at the 5% level of significance there is insufficient evidence to reject the company's claim. Although only 2 of the 50 tickets won, such an outcome is not unlikely enough (probability about 0.12 under H_0) to accuse the lottery company of overstating the winning probability. The association should collect a larger sample before complaining.

SOLUTION **Exercise 6 — The call centre**

1. The claimed rate is $\lambda = 6$ calls per hour; we suspect an increase:

$$H_0 : \lambda = 6 \qquad \text{against} \qquad H_1 : \lambda > 6 \quad \text{(one-tailed test).}$$

2. Under H_0, $X \sim P(6)$. From cumulative Poisson tables, $P(X \le 11) = 0.9799$, hence

$$P(X \ge 12) = 1 - P(X \le 11) = 1 - 0.9799 = 0.0201.$$

3. At the 1% level, we reject H_0 only if the p-value is below 0.01. Here

$$p\text{-value} = 0.0201 > 0.01,$$

so we do not reject H_0. Interpretation: observing 12 calls in an hour is unusual (probability about 2% when $\lambda = 6$), but not unlikely enough at the demanding 1% level to conclude that the mean rate has increased. The centre should monitor more hours before hiring extra staff. (Note: at the 5% level the same data would lead to rejection — the choice of α matters.)

SOLUTION **Exercise 7 — Sachet water**

1. We test whether the mean is below the claimed 500 ml:

$$H_0 : \mu = 500 \qquad \text{against} \qquad H_1 : \mu < 500 \quad \text{(one-tailed test).}$$

2. With $n = 36$ (large), $\bar{x} = 496$, $s = 9$, the sample mean is approximately normal by the Central Limit Theorem:

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{496 - 500}{9/\sqrt{36}} = \frac{-4}{9/6} = \frac{-4}{1.5} \approx -2.67.$$

The critical value for a one-tailed test at the 5% level is -1.645. Since $-2.67 < -1.645$, the statistic lies in the critical region: we reject H_0.

3. The p-value is

$$p = P(\bar{X} \le 496 \mid \mu = 500) = P(Z \le -2.67) = 1 - \Phi(2.67) \approx 1 - 0.9962 = 0.0038.$$

Since $0.0038 < 0.05 = \alpha$, the p-value approach confirms the rejection of H_0 (indeed the result is even significant at the 1% level).

4. Advice: there is strong evidence that the sachets are underfilled. The manager should stop the production line, recalibrate the filling machine, and check whether the underfilling is systematic, both to respect consumers and to avoid sanctions from the standards authorities.

SOLUTION **Exercise 8 — Bags of cement**

1. $H_0 : \mu = 50$ against $H_1 : \mu \neq 50$ (two-tailed). Here $\sigma = 1.2$ kg is known and $n = 25$:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{50.7 - 50}{1.2/\sqrt{25}} = \frac{0.7}{0.24} \approx 2.92.$$

   The critical values for a two-tailed test at the 5% level are ± 1.960. Since $2.92 > 1.960$, we reject H_0: there is significant evidence that the mean mass differs from 50 kg (in fact it appears to be higher).

2. The 95% confidence interval is

$$\bar{x} \pm 1.96\,\frac{\sigma}{\sqrt{n}} = 50.7 \pm 1.96 \times 0.24 = 50.7 \pm 0.47,$$

   i.e. $[50.23\,;\,51.17]$ kg. The claimed value 50 does not belong to this interval, so the confidence interval also leads to rejecting $H_0 : \mu = 50$ — the same conclusion as the test.

3. A 95% confidence interval consists precisely of all values μ_0 that would not be rejected by a two-tailed test at the 5% level, so the two procedures must always agree.

SOLUTION **Exercise 9 — Two schools at the mock GCE**

Indicative mark scheme: (a) 2 marks: H_0, H_1 with correct direction; (b) 2 marks: CLT quoted and applied to the difference; (c) 4 marks: variance formula (1), substitution (1), SE (1), z (1); (d) 3 marks: critical value, comparison, conclusion in context; (e) 1 mark: any valid assumption.

1. Let μ_A, μ_B be the true mean marks. School A is suspected to be higher:

$$H_0 : \mu_A = \mu_B \quad (\text{i.e. } \mu_A - \mu_B = 0) \qquad H_1 : \mu_A > \mu_B \quad (\text{one-tailed}).$$

2. The sample sizes $n_A = 60$ and $n_B = 80$ are both large. By the Central Limit Theorem, each sample mean is approximately normal, and the difference of two independent normal variables is normal:

$$\bar{X}_A - \bar{X}_B \approx N\!\left(\mu_A - \mu_B,\, \frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}\right).$$

3. Standard error:

$$\text{SE} = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = \sqrt{\frac{11.2^2}{60} + \frac{12.5^2}{80}} = \sqrt{\frac{125.44}{60} + \frac{156.25}{80}} = \sqrt{2.091 + 1.953} = \sqrt{4.044} \approx 2.01.$$

   Test statistic:

$$z = \frac{\bar{x}_A - \bar{x}_B - 0}{\text{SE}} = \frac{58.4 - 55.1}{2.01} = \frac{3.3}{2.01} \approx 1.64.$$

4. For a one-tailed test at the 5% level the critical value is 1.645. Since 1.64 < 1.645, the statistic does not fall in the critical region (the p-value is about 0.051). We do not reject H_0: at the 5% level there is insufficient evidence that school A's mean mark is significantly higher than school B's — the observed gap of 3.3 marks can plausibly be due to sampling variation.

5. Assumptions (any one): the two samples are independent random samples; the sample standard deviations are reliable estimates of the population standard deviations (justified by the large samples); marks within each school are independent of one another.

Examiner's expectations: candidates must write hypotheses in terms of population means, quote the CLT by name, use the unpooled standard error $\sqrt{s_A^2/n_A + s_B^2/n_B}$, and end with a conclusion in context (mentioning the schools), not a bare "accept H_0".

SOLUTION **Exercise 10 — Finding a critical region**

Indicative mark scheme: (a) 2 marks; (b) 4 marks (method 1, each probability 1.5); (c) 3 marks (choice 2, actual level 1); (d) 1 mark.

1. Under $H_1 : p < 0.4$, the number of germinating seeds will tend to be smaller than expected under H_0 (where $E(X) = 20 \times 0.4 = 8$). Small values of X are therefore the evidence against H_0, so the critical region consists of the smallest values of X, i.e. has the form $X \le c$.

2. With $X \sim B(20, 0.4)$, $P(X = k) = \binom{20}{k}(0.4)^k(0.6)^{20-k}$:

$$P(X \le 4) = \sum_{k=0}^{4} \binom{20}{k}(0.4)^k(0.6)^{20-k}$$
$$= 0.00004 + 0.00049 + 0.00309 + 0.01235 + 0.03499 \approx 0.0510,$$
$$P(X \le 3) = 0.00004 + 0.00049 + 0.00309 + 0.01235 \approx 0.0160.$$

3. We want the level as close as possible to 5%: $|0.0510 - 0.05| = 0.001$ while $|0.0160 - 0.05| = 0.034$. Hence the critical region is

$$X \le 4, \qquad \text{with actual significance level } \alpha = P(X \le 4 \mid p = 0.4) \approx 5.1\%.$$

(Because X is discrete, an exact 5% level is unattainable — this is a classic GCE point.)

4. The buyer observes $X = 4$, which lies in the critical region $X \le 4$. He therefore rejects H_0 at (approximately) the 5% level: the data support his suspicion that the germination rate is below 0.4.

SOLUTION **Exercise 11 — Malaria cases after a flood**

Indicative mark scheme: (a) 3 marks (hypotheses 2, justification 1); (b) 4 marks; (c) 3 marks; (d) 2 marks; (e) 2 marks.

1. The historical mean is $\lambda = 3$ cases per day; the officer suspects an increase after the flood. Since only larger counts would indicate a worsening situation, a one-tailed test is appropriate:
$$H_0 : \lambda = 3 \qquad \text{against} \qquad H_1 : \lambda > 3.$$

2. Under H_0, $X \sim P(3)$. From cumulative Poisson tables, $P(X \le 8) = 0.9962$, so the p-value is
$$P(X \ge 9 \mid \lambda = 3) = 1 - P(X \le 8) = 1 - 0.9962 = 0.0038.$$

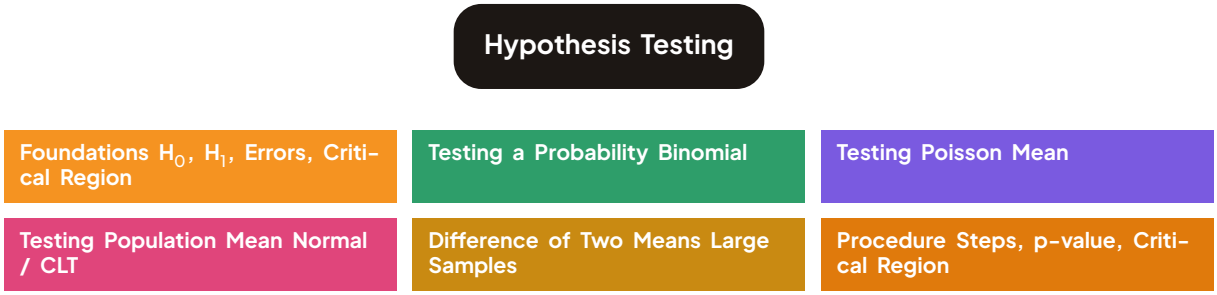
3. At the 1% level of significance we reject H_0 if the p-value is below 0.01. Since
$$0.0038 < 0.01,$$
we reject H_0 at the 1% level.

4. Conclusion to the medical officer: if the mean rate were still 3 cases per day, a day with 9 or more cases would occur with probability less than 0.4% — an extremely rare event. There is therefore strong evidence (significant even at the 1% level) that the mean daily number of malaria cases has increased after the flood. Preventive measures (distribution of treated nets, drainage of stagnant water, extra stock of antimalarials) should be reinforced immediately.

5. Since we rejected H_0, the only possible error is a Type I error: it would mean that the mean rate has in fact not increased, and the clinic would be mobilising resources and alarm unnecessarily for a surge that was purely due to chance.

Examiner's expectations: hypotheses stated in terms of λ with the context named; the p-value computed as a tail probability $P(X \ge 9)$ and not $P(X = 9)$ (a very common mistake); an explicit comparison with 0.01; a contextual conclusion; correct identification of the error type consistent with the decision taken.

13 SUMMARY SHEET



Mind map of Chapter FMA21: Hypothesis Testing

FICHE BILAN **L'essentiel à connaître pour l'examen**

13.1 Key Definitions

- **Null hypothesis (H_0)**: Statement assumed true unless evidence contradicts it (e.g. $p = p_0, \mu = \mu_0, \lambda = \lambda_0, \mu_1 - \mu_2 = 0$).
- **Alternative hypothesis (H_1)**: Statement accepted if H_0 is rejected; can be one-tailed ($<$, $>$) or two-tailed ($\neq$).
- **Significance level (α)**: Probability of Type I error (rejecting H_0 when true); common values 0.05, 0.01, 0.10.
- **Critical region**: Set of test statistic values leading to rejection of H_0.
- **p-value**: Probability, assuming H_0 true, of obtaining a result at least as extreme as the observed one.
- **Type I error**: Reject H_0 when H_0 is true (α).
- **Type II error**: Fail to reject H_0 when H_0 is false (β).
- **Power of test**: $1 - \beta$ = probability of correctly rejecting a false H_0.

13.2 Laws, Formulas & Theorems to Know

- Test statistic for a probability (Binomial): $X \sim B(n, p_0)$. For large n, use normal approximation $Z = \dfrac{X - np_0}{\sqrt{np_0(1-p_0)}} \sim N(0,1)$ (with continuity correction).
- Test statistic for Poisson mean: $X \sim Po(\lambda_0)$. For large λ_0, $Z = \dfrac{X - \lambda_0}{\sqrt{\lambda_0}} \sim N(0,1)$ (continuity

correction). Exact test uses Poisson tables.

- Test statistic for population mean (σ known): $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$.
- Test statistic for population mean (σ unknown, large n): $Z = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim N(0,1)$ by CLT ($n \geq 30$).
- Test statistic for difference of two means (large samples):

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

  If σ_i unknown, replace by sample standard deviations S_i ($n_i \geq 30$).
- Critical values: z_α (one-tailed), $z_{\alpha/2}$ (two-tailed) from standard normal tables.

13.3 Standard Method (Steps for Every Test)

1. State H_0 and H_1 clearly, defining parameters.
2. Choose significance level α (given or 5% default).
3. Identify the appropriate test statistic and its distribution under H_0.
4. Determine critical region or compute p-value.
5. Calculate the test statistic value from sample data.
6. Make decision: reject H_0 if test statistic falls in critical region or p-value $< \alpha$; otherwise do not reject.
7. Write a conclusion in context of the problem.

13.4 Common Mistakes & Traps

- Confusing one-tailed and two-tailed critical regions (e.g. using $z_{0.05} = 1.645$ instead of $z_{0.025} = 1.96$ for two-tailed 5%).
- Forgetting continuity correction when approximating discrete distributions (Binomial, Poisson) by Normal.
- Using t-test instead of Z-test for large samples ($n \geq 30$); syllabus requires Normal/CLT.
- Stating "accept H_0" instead of "do not reject H_0".
- Misidentifying the parameter: testing p vs λ vs μ vs $\mu_1 - \mu_2$.

- Using sample standard deviation S in denominator without checking $n \geq 30$ (or population normal with known σ).
- Not checking conditions: $np_0 > 5$, $n(1 - p_0) > 5$ for Binomial approximation; $\lambda_0 > 10$ for Poisson approximation.

13.5 GCE Examination Tips

- Always define the random variable and its distribution under H_0 before computing.
- Show the formula for the test statistic before substituting numbers.
- For two-tailed tests, halve α for critical values; double the one-tail probability for p-value.
- When using normal approximation, write the continuity correction explicitly (e.g. $P(X \geq 12) \approx P(Z \geq \frac{11.5 - np}{\sqrt{npq}})$).
- If the question asks "test at the 5% level", state the critical value(s) from tables ($z_{0.05} = 1.645$, $z_{0.025} = 1.96$).
- For difference of two means, ensure samples are independent and large enough ($n_1, n_2 \geq 30$).
- Final conclusion must be non-assertive: "There is sufficient evidence at the 5% level to suggest that…" or "There is insufficient evidence…".
- Manage time: a full hypothesis test carries 6–8 marks; do not skip steps.

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